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Yamaga

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(54) **MAGNETIC RECORDING MEDIUM AND CARTRIDGE**

(71) Applicant: **Sony Group Corporation**, Tokyo (JP)

(72) Inventor: **Minoru Yamaga**, Tokyo (JP)

(73) Assignee: **Sony Group Corporation**, Tokyo (JP)

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None

See application file for complete search history.

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Primary Examiner — Fred Tzeng

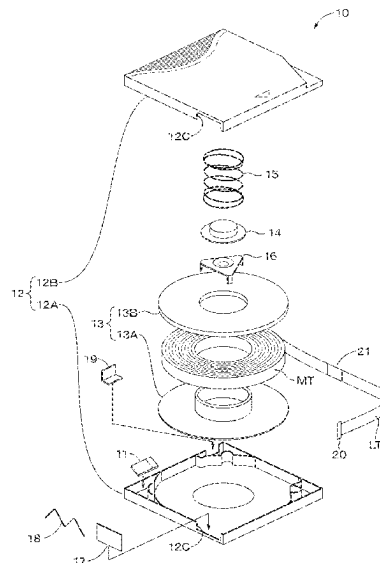
(74) *Attorney, Agent, or Firm* — K&L Gates LLP

(57)

ABSTRACT

A magnetic recording medium is a tape-shaped magnetic recording medium and includes: a base; and a magnetic layer. An average thickness of the magnetic recording medium is 5.30 μm or less. The magnetic layer contains a chlorine atom and a nitrogen atom, an atomic concentration ratio (A2/A1) of a maximum chlorine atomic concentration A1 obtained by measuring a chlorine atomic concentration of the magnetic layer in a thickness direction of the magnetic layer and a maximum nitrogen atomic concentration A2 obtained by measuring a nitrogen atomic concentration of the magnetic layer in the thickness direction of the magnetic layer is 0.50 or more and 0.75 or less. An average value of a protruding peak height Rpk of a surface of the magnetic layer is 2.10 nm or less.

16 Claims, 16 Drawing Sheets



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	<i>G11B 23/037</i>		(2006.01)						428/141	
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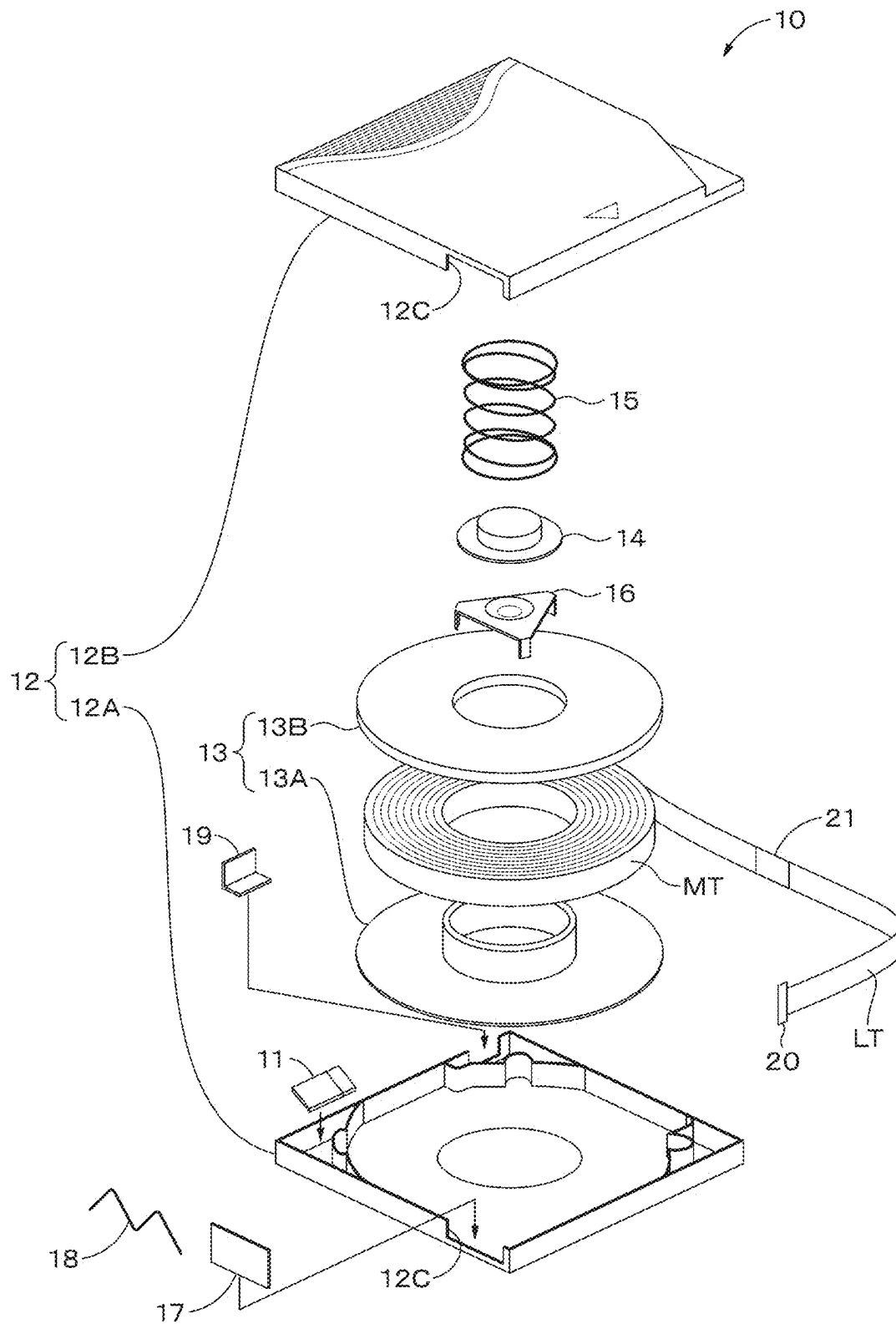


FIG. 1

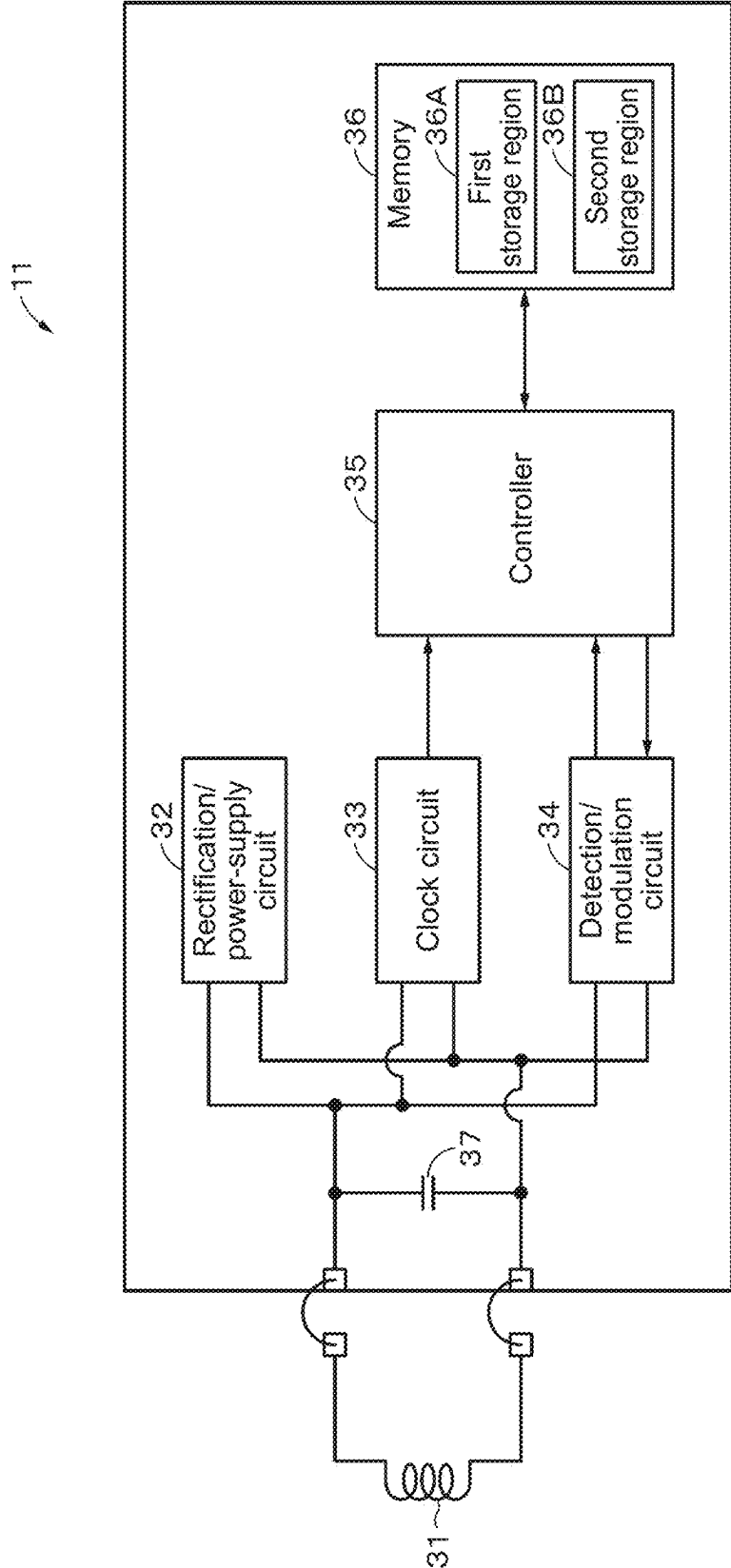


FIG.2

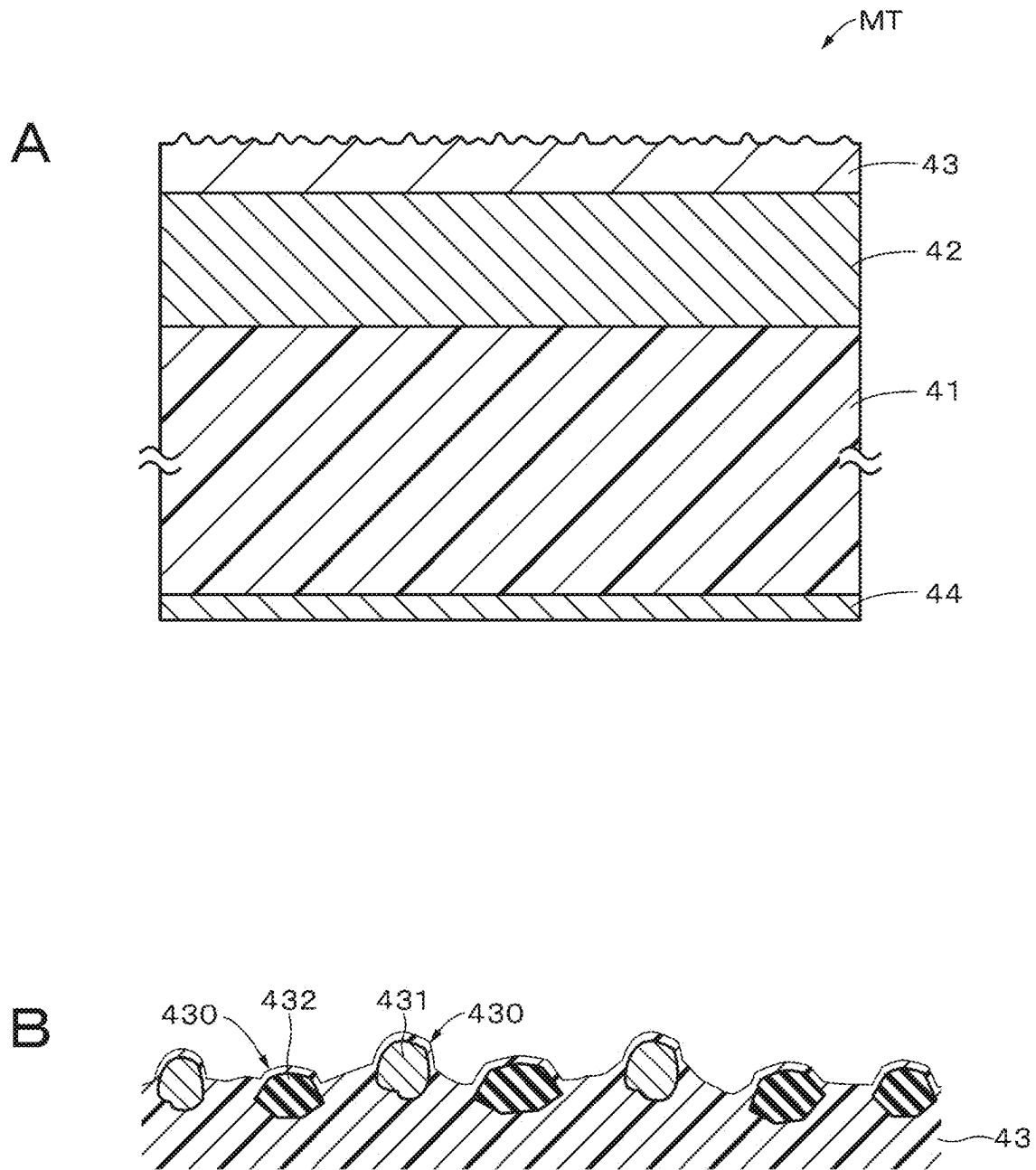


FIG.3

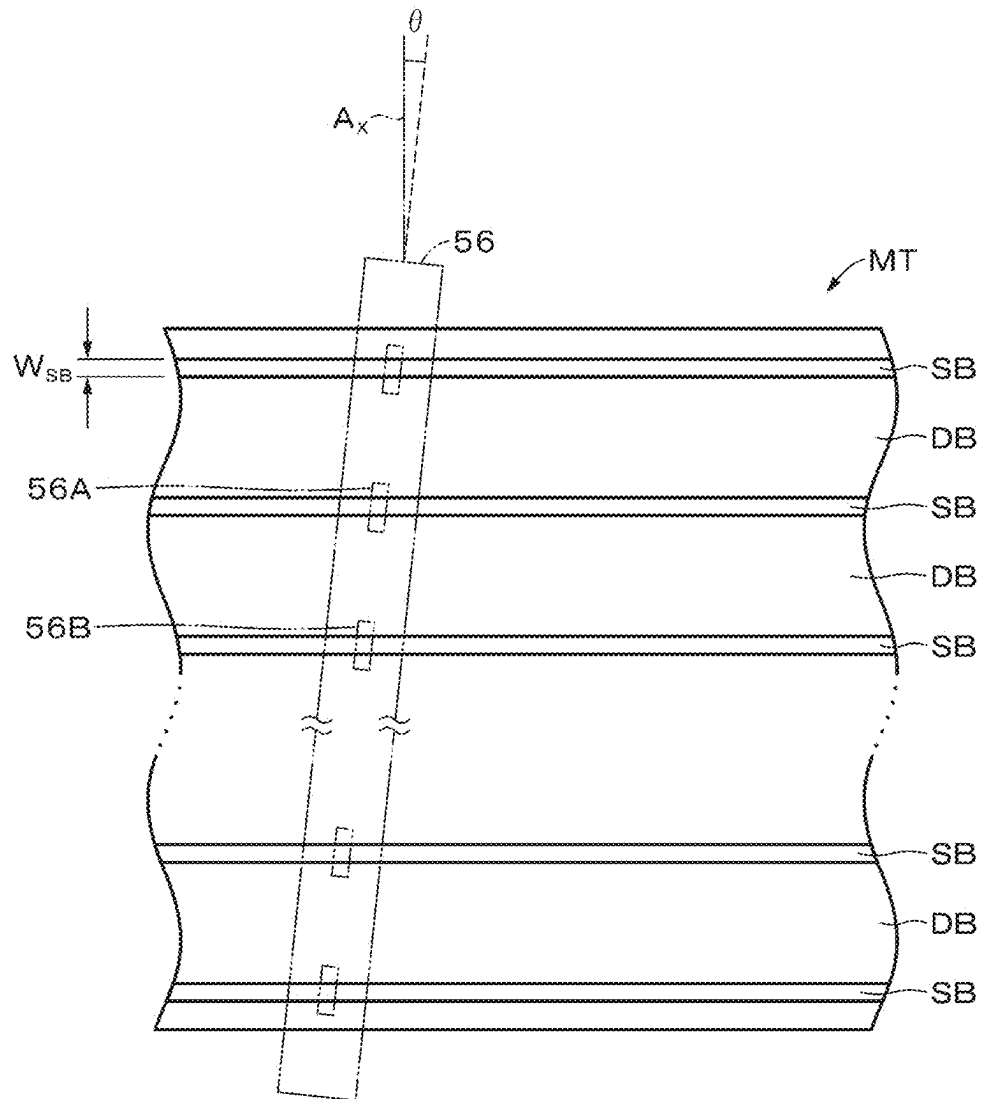


FIG. 4

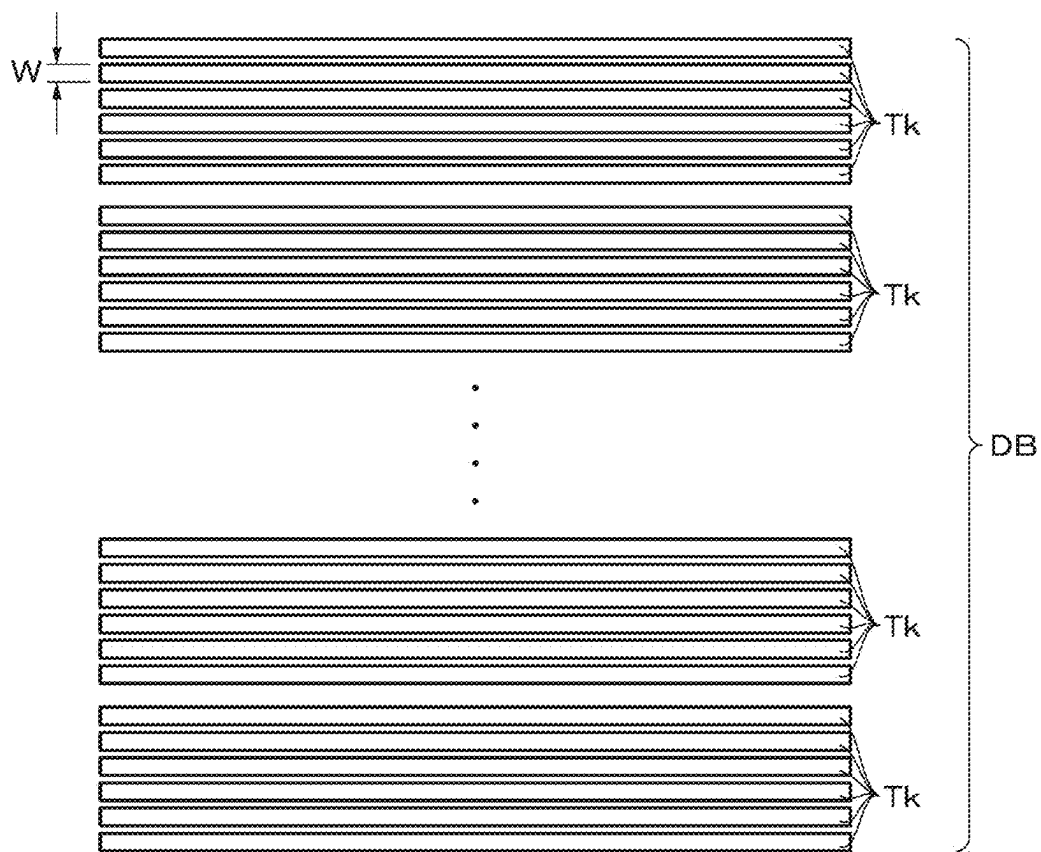


FIG.5

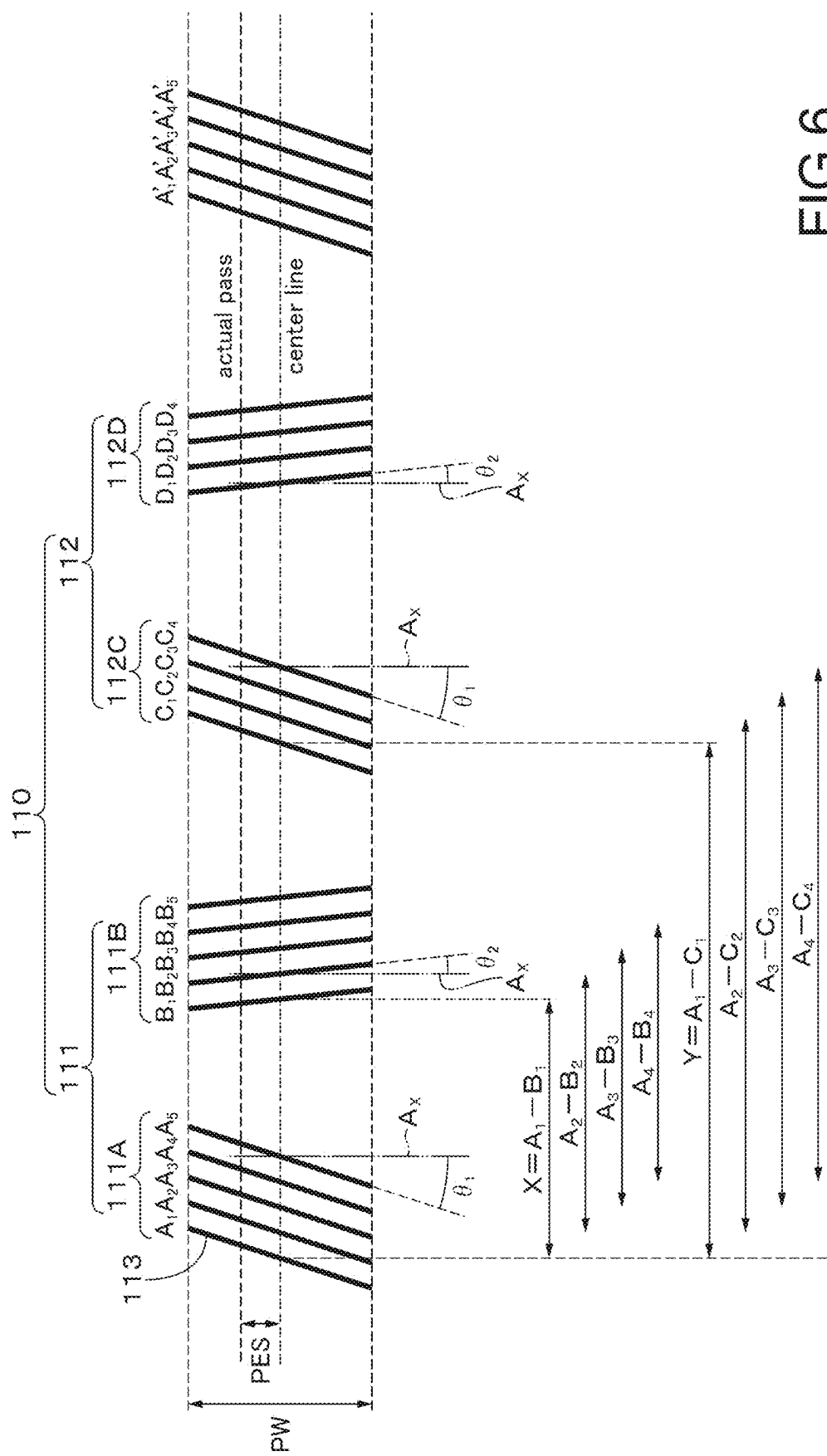


FIG.6

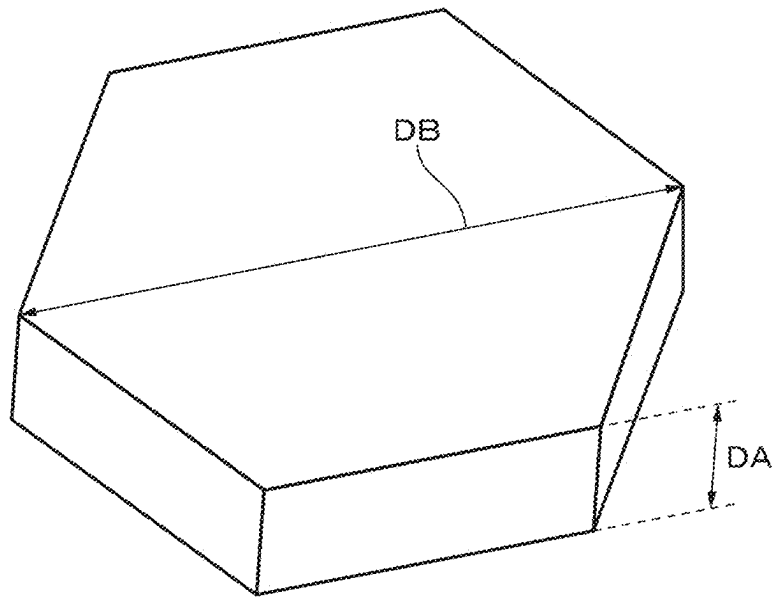


FIG.7

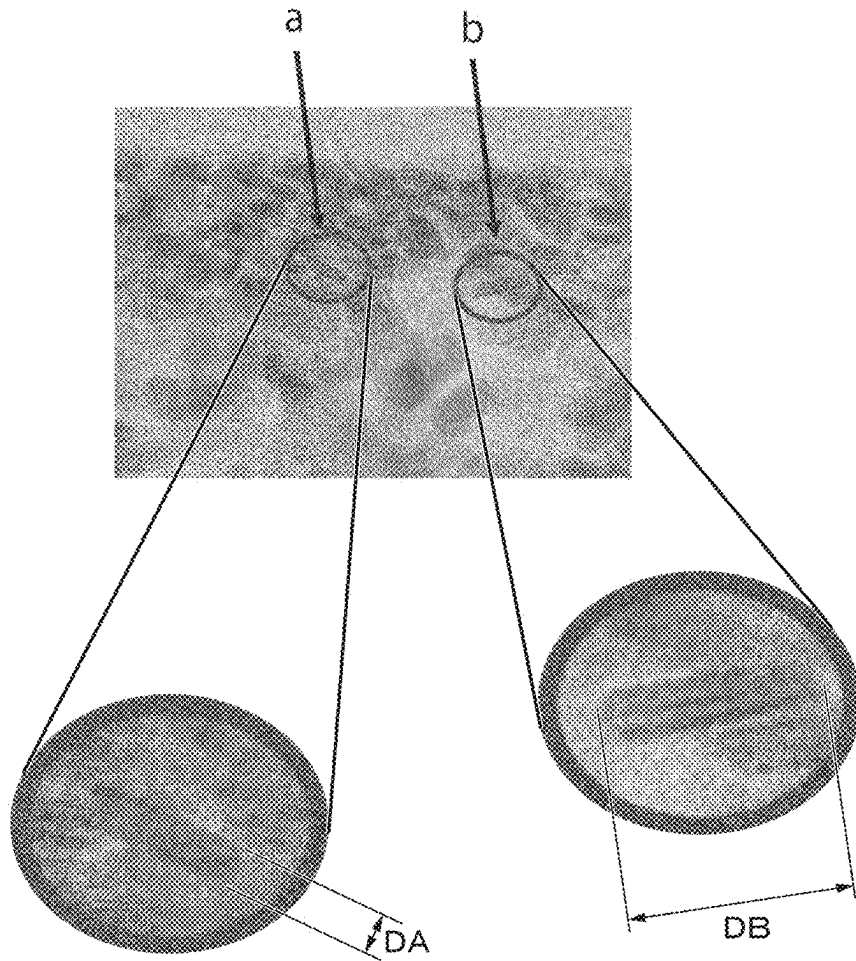


FIG.8

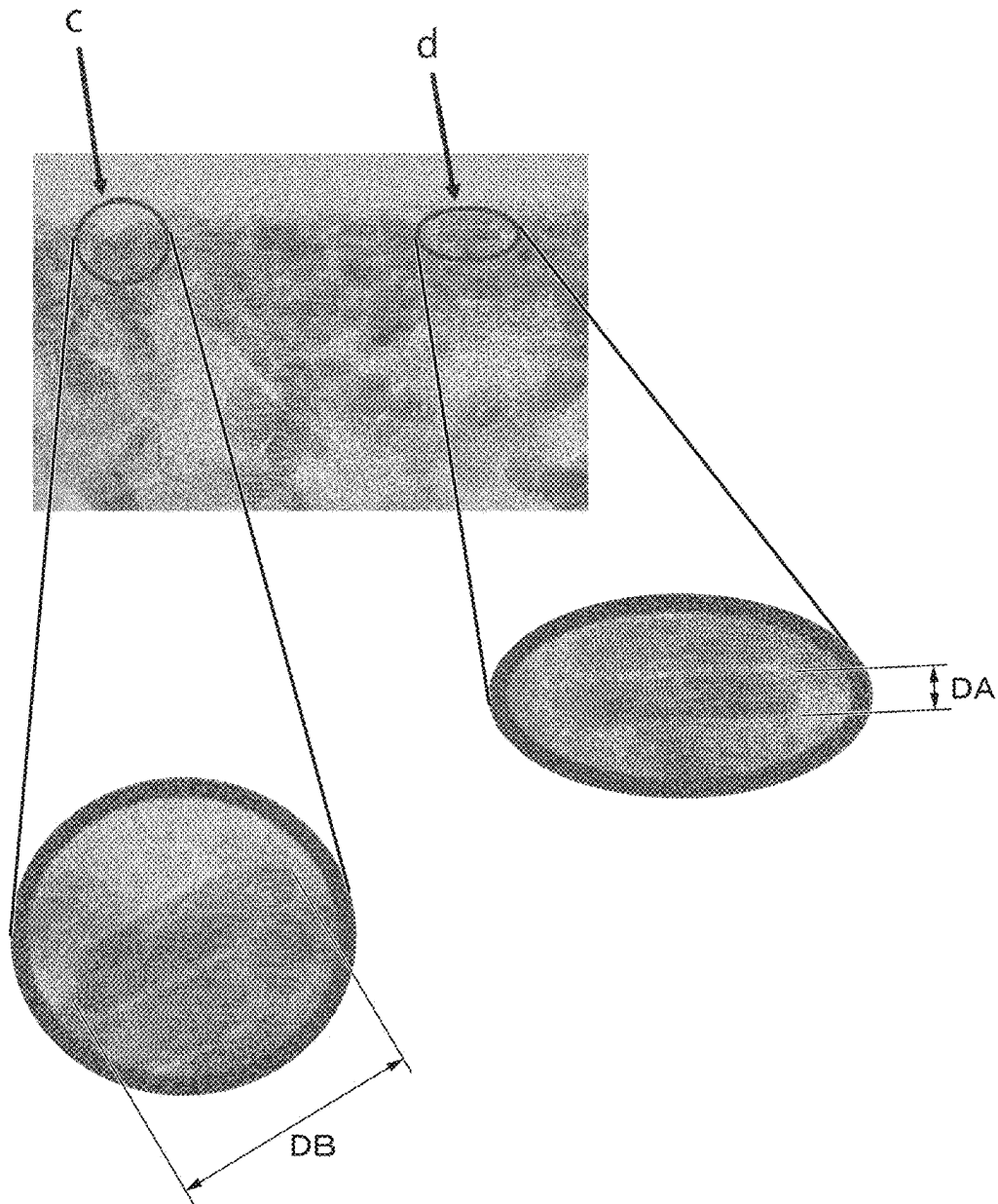


FIG.9

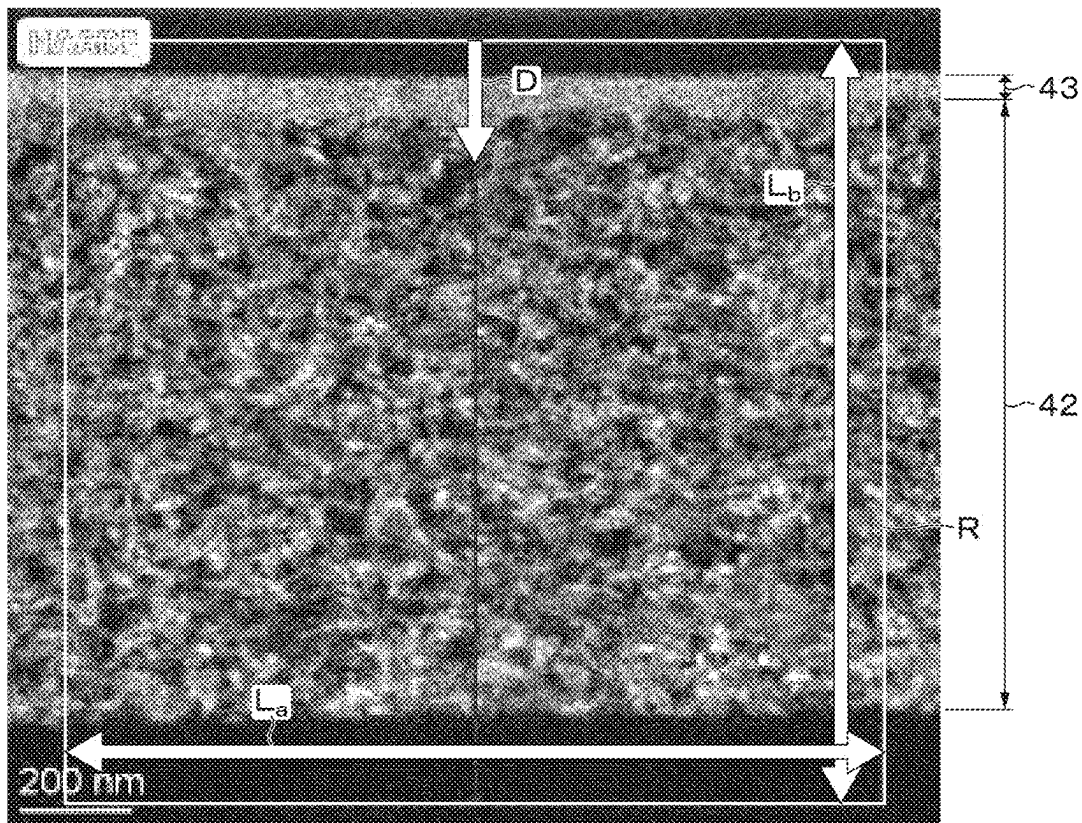


FIG.10

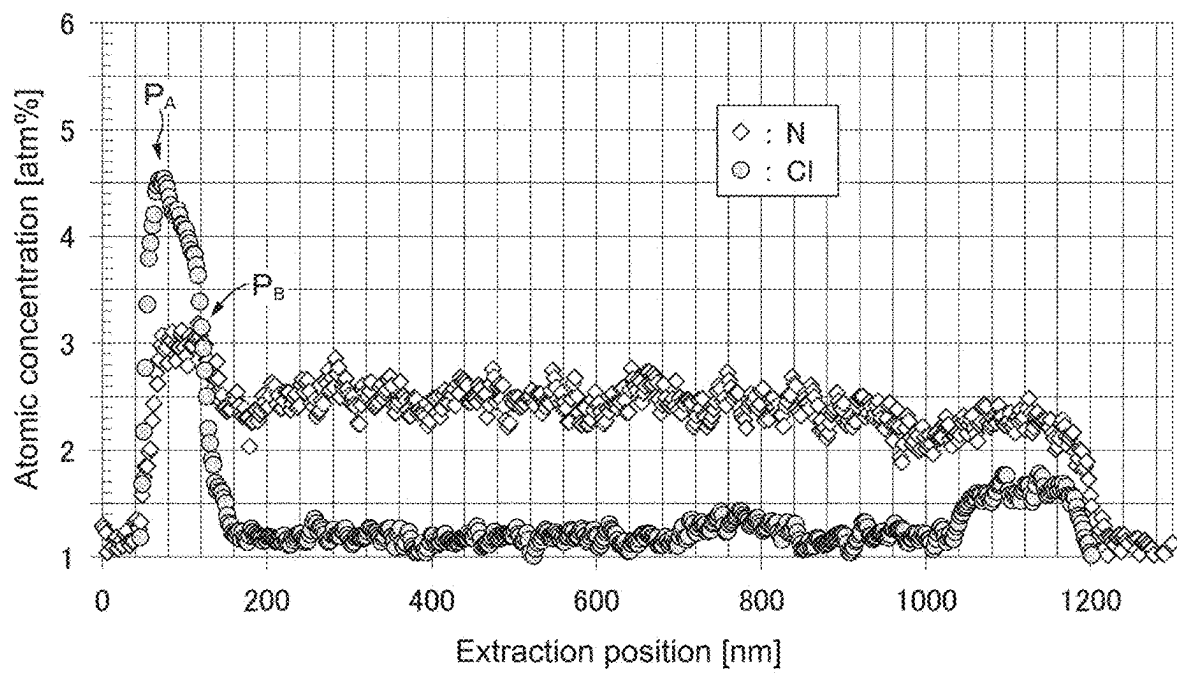
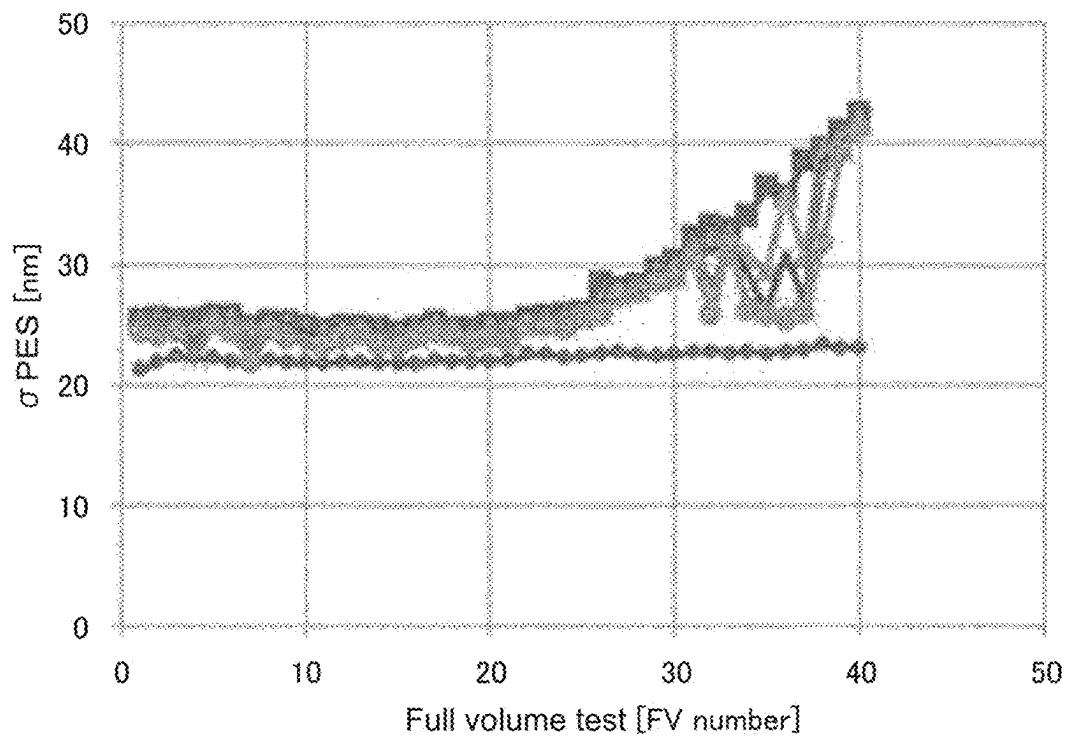
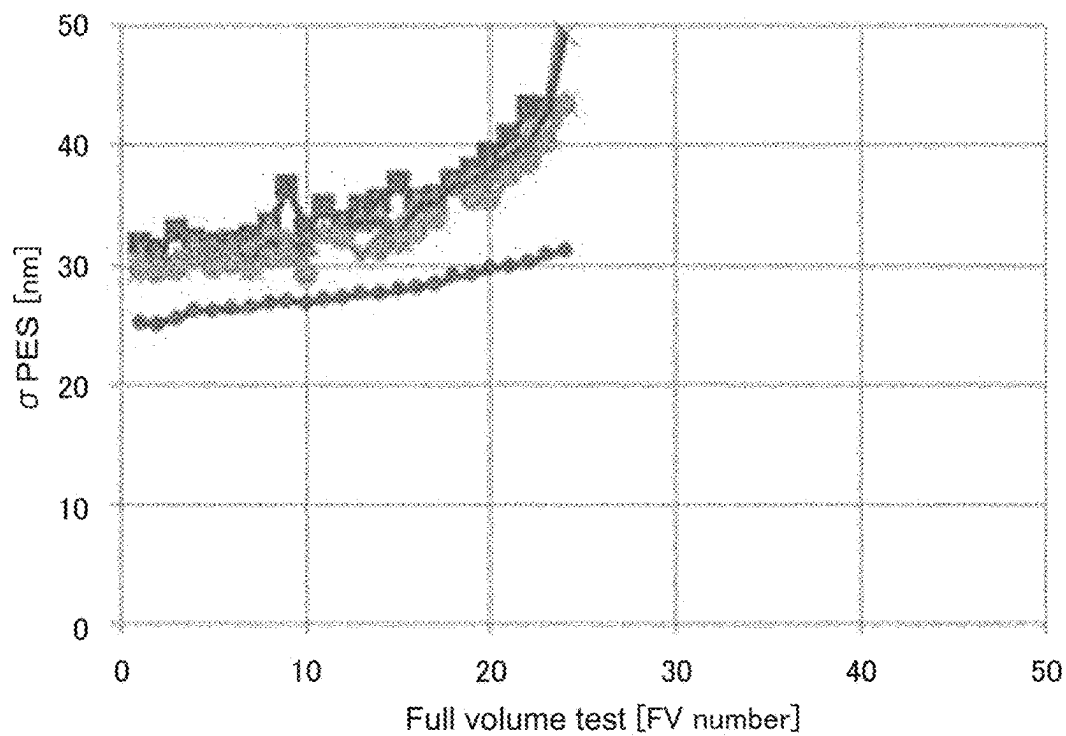


FIG.11

A**B****FIG.12**

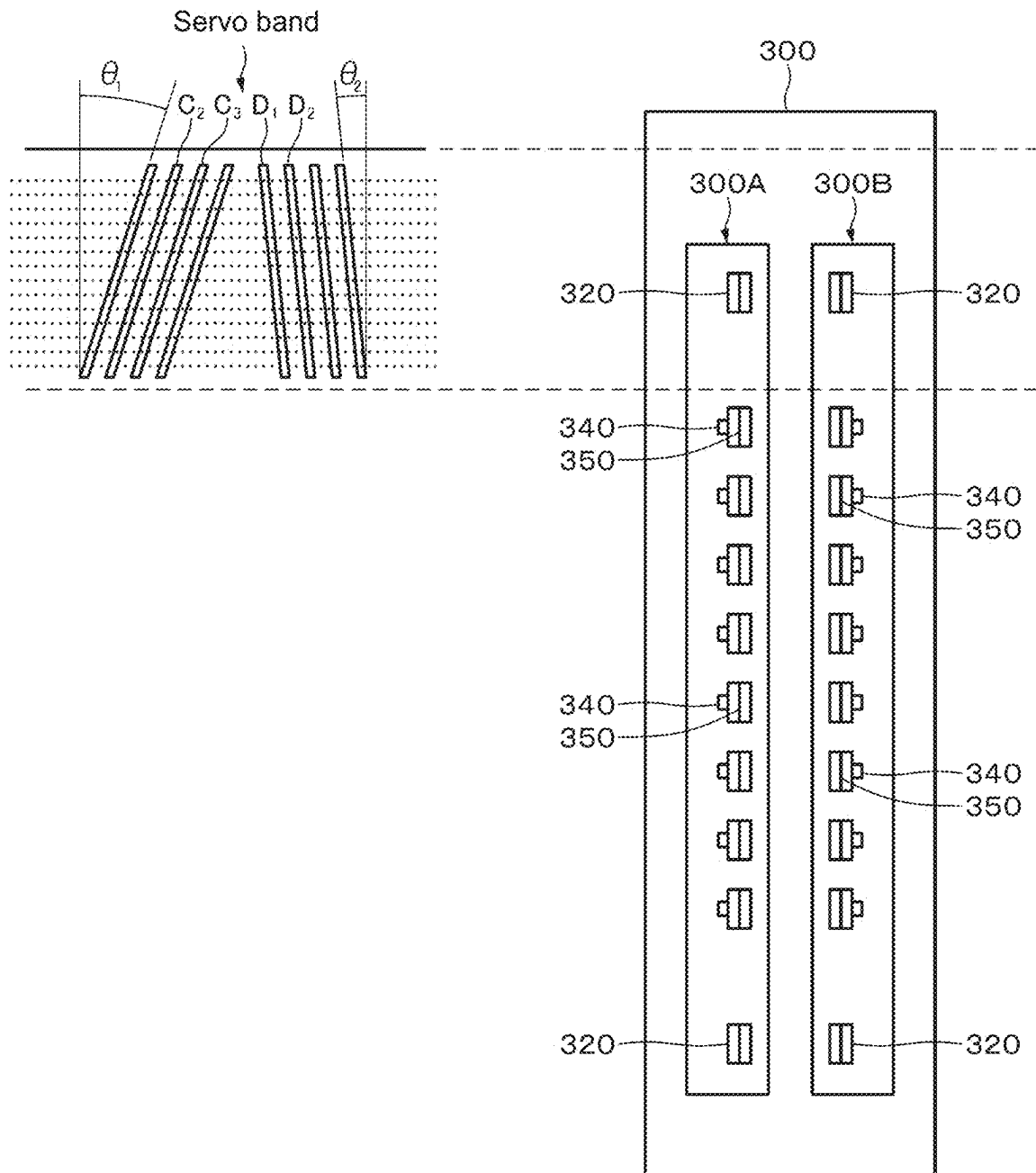


FIG.13

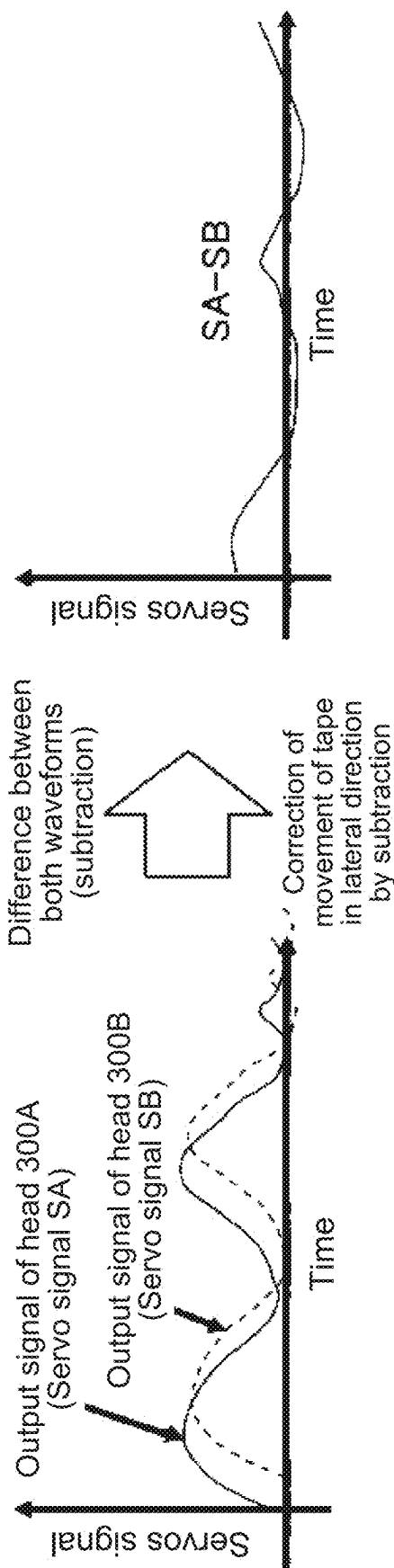


FIG.14

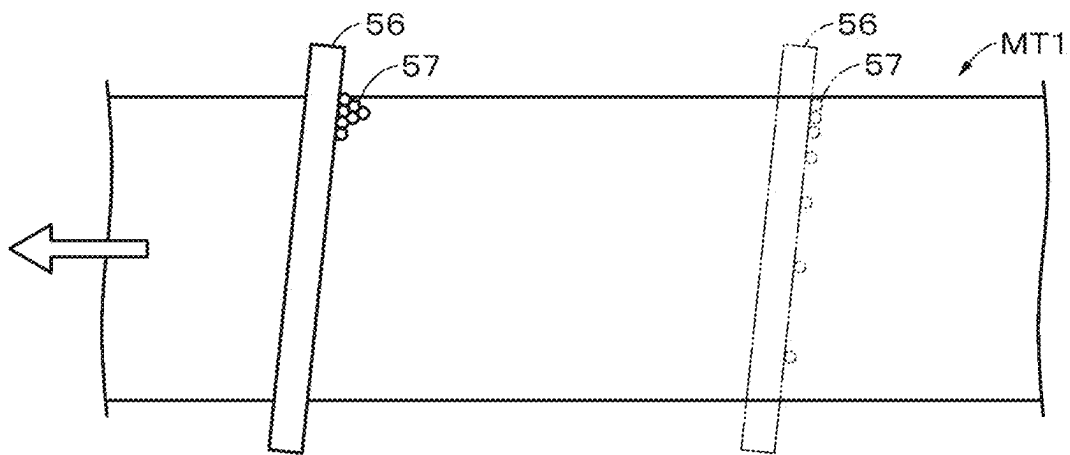


FIG.15

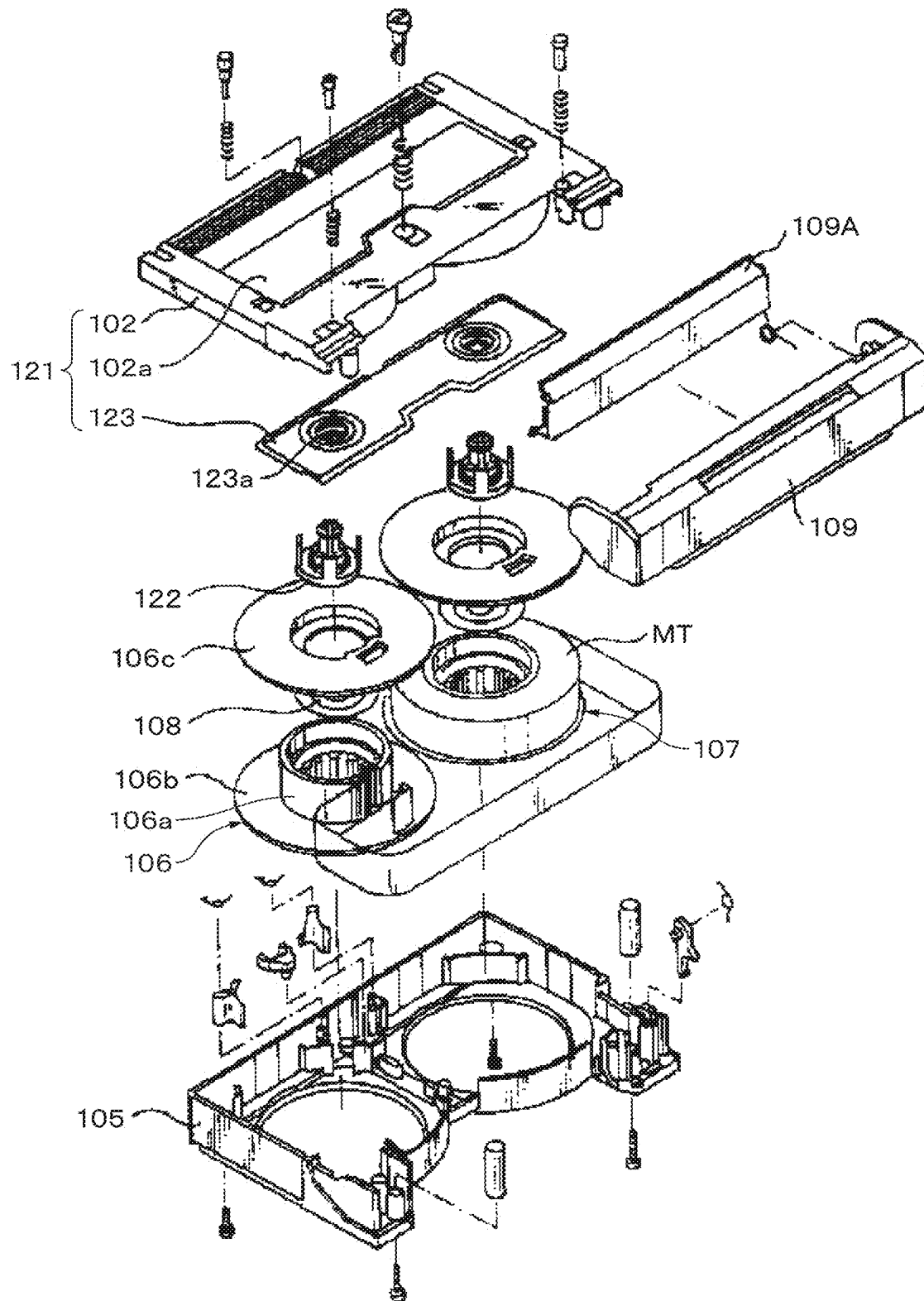


FIG. 16

MAGNETIC RECORDING MEDIUM AND CARTRIDGE

TECHNICAL FIELD

The present disclosure relates to a magnetic recording medium and a cartridge including the same.

BACKGROUND ART

Tape-shaped magnetic recording media are widely used to store electronic data. In the tape-shaped magnetic recording media, in order to achieve favorable recording and reproduction characteristics (electromagnetic conversion characteristics), it is desired to reduce the height of recesses and projections on the surface of the magnetic layer and smooth the surface of the magnetic layer (see, for example, Patent Literature 1).

CITATION LIST

Patent Literature

Patent Literature 1: Japanese Patent Application Laid-open No. 2006-65953

DISCLOSURE OF INVENTION

Technical Problem

However, there is a possibility that reducing the recesses and projections on the surface of the magnetic layer leads to an increase in friction between the magnetic tape and the head unit.

It is an object of the present disclosure to provide a magnetic recording medium that is capable of achieving favorable electromagnetic conversion characteristics and suppressing the increase in friction and a cartridge including the same.

Solution to Problem

In order to achieve the above-mentioned object, a magnetic recording medium according to the present disclosure is

- a tape-shaped magnetic recording medium, including:
- a base; and
- a magnetic layer,
- an average thickness of the magnetic recording medium being 5.30 μm or less,
- the magnetic layer containing a chlorine atom and a nitrogen atom,
- an atomic concentration ratio ($A2/A1$) of a maximum chlorine atomic concentration $A1$ obtained by measuring a chlorine atomic concentration of the magnetic layer in a thickness direction of the magnetic layer and a maximum nitrogen atomic concentration $A2$ obtained by measuring a nitrogen atomic concentration of the magnetic layer in the thickness direction of the magnetic layer being 0.50 or more and 0.75 or less,
- an average value of a protruding peak height R_{pk} of a surface of the magnetic layer being 2.10 nm or less.

A cartridge according to the present disclosure includes the magnetic recording medium according to the present disclosure.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is an exploded perspective view showing an example of a configuration of a cartridge according to an embodiment of the present disclosure.

FIG. 2 is a block diagram showing an example of a configuration of a cartridge memory.

Part A of FIG. 3 is a cross-sectional view showing an example of a configuration of a magnetic tape. Part B of FIG. 3 is a cross-sectional view showing an example of a configuration of a surface of the magnetic layer.

FIG. 4 is a schematic diagram showing an example of the layout of a data band and a servo band.

FIG. 5 is an enlarged view showing an example of a configuration of the data band.

FIG. 6 is an enlarged view showing an example of a configuration of the servo band.

FIG. 7 is a perspective view showing an example of a shape of a particle.

FIG. 8 is a diagram showing a first example of a cross-sectional TEM image of the magnetic layer.

FIG. 9 is a diagram showing a second example of the cross-sectional TEM image of the magnetic layer.

FIG. 10 is a diagram showing an example of a HAADF STEM image.

FIG. 11 is a graph showing an example of atomic concentration distribution of chlorine and nitrogen in a depth direction (thickness direction).

Part A of FIG. 12 is a graph showing a first example of the change in standard deviation σPES over time. Part B of FIG. 12 is a graph showing a second example of the change in standard deviation σPES over time.

FIG. 13 is a diagram describing a method of measuring PES.

FIG. 14 is a graph describing correction of movement in a width direction of the magnetic tape.

FIG. 15 is a diagram showing how a powder generated by scraping off a protrusion spreads.

FIG. 16 is an exploded perspective view showing an example of a configuration of a cartridge according to a modified example of the embodiment of the present disclosure.

MODE(S) FOR CARRYING OUT THE INVENTION

An embodiment of the present disclosure will be described in the following order.

- 1 Configuration of cartridge
- 2 Configuration of cartridge memory
- 3 Configuration of magnetic tape
- 4 Method of producing magnetic tape
- 5 Operation and effect
- 6 Modified example

In the present specification, in the case where a measurement environment is not particularly described regarding description of a measurement method, the measurement is performed in an environment of 25° C. $\pm$ 2° C. and 50% RH $\pm$ 5% RH.

1 Configuration of Cartridge

FIG. 1 is an exploded perspective view showing an example of a configuration of the cartridge 10. The cartridge 10 is a one-reel type cartridge and includes, inside a cartridge case 12 including a lower shell 12A and an upper shell 12B, one reel 13 on which a tape-shaped magnetic recording

medium (hereinafter, referred to as a “magnetic tape”.) MT is wound, a reel lock **14** and a reel spring **15** for locking the rotation of the reel **13**, a spider **16** for releasing the locked state of the reel **13**, a slide door **17** for opening and closing a tape outlet **12C** provided in the cartridge case **12** across the lower shell **12A** and the upper shell **12B**, a door spring **18** that biases the slide door **17** to the closed position of the tape outlet **12C**, a write protector for preventing accidental erasure, and a cartridge memory **11**. The reel **13** for winding a magnetic tape MT has a substantially disk shape with an opening in the center and includes a reel hub **13A** and a flange **13B** formed of a hard material such as plastic. A leader tape LT is connected to an end portion of the magnetic tape MT on the outer periphery side. A leader pin **20** is provided at the tip of the leader tape LT.

The cartridge **10** may be a magnetic tape cartridge conforming to the LTO (Linear Tape-Open) standard or may be magnetic tape cartridge conforming to a standard other than the LTO standard.

The cartridge memory **11** is provided in the vicinity of one corner portion of the cartridge **10**. When the cartridge **10** is loaded into a recording/reproduction apparatus, the cartridge memory **11** faces the reader/writer of the recording/reproduction apparatus. The cartridge memory **11** communicates with the recording/reproduction apparatus, specifically the reader/writer, using a wireless communication standard conforming to the LTO standard.

2 Configuration of Cartridge Memory

FIG. **2** is a block diagram showing an example of a configuration of the cartridge memory **11**. The cartridge memory **11** includes an antenna coil (communication unit) **31** that communicates with a reader/writer using a predetermined communication standard, a rectification/power-supply circuit **32** that generates power from radio waves received by the antenna coil **31** using an induced electromotive force and rectifies the power to generate power supply, a clock circuit **33** that generates a clock from radio waves received by the antenna coil **31** using an induced electromotive force similarly, a detection/modulation circuit **34** that detects radio waves received by the antenna coil **31** and modulates signals to be transmitted by the antenna coil **31**, a controller (control unit) **35** that includes a logic circuit or the like for determining and processing a command and data from a digital signal extracted from the detection/modulation circuit **34**, and a memory (storage unit) **36** that stores information. Further, the cartridge memory **11** includes a capacitor **37** connected in parallel to the antenna coil **31**, and the antenna coil **31** and the capacitor **37** constitute a resonant circuit.

The memory **36** stores information relating to the cartridge **10**, and the like. The memory **36** is a non-volatile memory (NVM). The memory capacity of the memory **36** is favorably approximately 32 KB or more.

The memory **36** may have a first storage region **36A** and a second storage region **36B**. The first storage region **36A** corresponds to, for example, the storage region of the cartridge memory in a magnetic tape standard before a predetermined generation (e.g., the LTO standard before LTO8) and is a region for storing information conforming to the magnetic tape standard before the predetermined generation. The information conforming to the magnetic tape standard before the predetermined generation is, for example, manufacturing information (e.g., the unique number of the cartridge **10**) or a usage history (e.g., the number of times the tape has been pulled out (Thread Count)).

The second storage region **36B** corresponds to the expanded storage region for the storage region of the cartridge memory in the magnetic tape standard before the predetermined generation (e.g., the LTO standard before LTO8). The second storage region **36B** is a region for storing additional information. Here, the additional information means, for example, information relating to the cartridge **10**, which is not specified in the magnetic tape standard before the predetermined generation (e.g., the LTO standard before LTO8). The additional information includes, for example, at least one type of information selected from the group consisting of tension adjustment information, management ledger data, Index information, and thumbnail information, but is not limited these pieces of data. The tension adjustment information is information for adjusting the tension applied in the longitudinal direction of the magnetic tape MT. The tension adjustment information includes, for example, at least one type of information selected from the group consisting of information obtained by intermittently measuring the width between servo bands in the longitudinal direction of the magnetic tape MT, tension information of a drive, and information regarding the temperature and humidity of the drive. These pieces of information are managed in conjunction with information regarding the usage status of the cartridge **10** in some cases. The tension adjustment information is favorably obtained when data is recorded on the magnetic tape MT or before data is recorded. The information of the drive means information regarding the tension applied in the longitudinal direction of the magnetic tape MT.

The management ledger data is data that includes at least one selected from the group consisting of the capacity, creation date, editing date, and storage location of the data file recorded on the magnetic tape MT. The Index information includes metadata for searching for the content of the data file, or the like. The thumbnail information includes the thumbnail of a moving image or a still image stored in the magnetic tape MT.

The memory **36** may include a plurality of banks. In this case, some of the plurality of banks may form the first storage region **36A** and the remaining banks may form the second storage region **36B**.

The antenna coil **31** induces an induced voltage by electromagnetic induction. The controller **35** communicates with a recording/reproduction apparatus in accordance with the predetermined communication standard via the antenna coil **31**. Specifically, for example, mutual authentication, transmission/reception of a command, data exchange, or the like is performed.

The controller **35** stores information received from the recording/reproduction apparatus via the antenna coil **31** in the memory **36**. For example, tension adjustment information received from the recording/reproduction apparatus via the antenna coil **31** is stored in the second storage region **36B** of the memory **36**. The controller **35** reads information from the memory **36** in accordance with a request from the recording/reproduction apparatus, and transmits the read information to the recording/reproduction apparatus via the antenna coil **31**. For example, in accordance with a request from the recording/reproduction apparatus, tension adjustment information is read from the second storage region **36B** of the memory **36**, and the read tension adjustment information is transmitted to the recording/reproduction apparatus via the antenna coil **31**.

3 Configuration of Magnetic Tape

Part A of FIG. **3** is a cross-sectional view showing an example of a configuration of the magnetic tape MT. The

magnetic tape MT includes a long base **41**, an underlayer **42** provided on one main surface (first main surface) of the base **41**, a magnetic layer **43** provided on the underlayer **42**, and a back layer **44** provided on the other main surface (second main surface) of the base **41**. Note that the underlayer **42** and the back layer **44** are provided as necessary and may be omitted. The magnetic tape MT may be a perpendicular recording magnetic recording medium or a longitudinal recording magnetic recording medium. From the viewpoint of improving travelling performance, the magnetic tape MT favorably contains a lubricant. The lubricant may be contained in at least one of the underlayer **42** and the magnetic layer **43**.

The magnetic tape MT may be one conforming to the LTO standard or may be one conforming to a standard other than the LTO standard. The width of the magnetic tape MT may be ½ inch or may be wider than ½ inch. In the case where the magnetic tape MT conforms to the LTO standard, the width of the magnetic tape MT is ½ inch. The magnetic tape MT may have a configuration in which the width of the magnetic tape MT can be kept constant or substantially constant by adjusting the tension applied in the longitudinal direction of the magnetic tape MT during by the recording/reproduction apparatus (drive).

The magnetic tape MT has a long shape and is caused to travel in the longitudinal direction during recording and reproduction. The magnetic tape MT is favorably used in a recording/reproduction apparatus that includes a ring-type head as a recording head. The magnetic tape MT is favorably used in a recording/reproduction apparatus that is configured to be capable of recording data with a data track width of 1500 nm or less or 1000 nm or less.

The magnetic tape MT is favorably reproduced by a reproduction head using a TMR element. The signal to be reproduced by the reproduction head using TMR may be data recorded in a data band DB (see FIG. 4) or a servo pattern (servo signal) recorded in the servo band SB (see FIG. 4).

(Base)

The base **41** is a non-magnetic support that supports the underlayer **42** and the magnetic layer **43**. The base **41** has a long film shape. The upper limit value of the average thickness of the base **41** is favorably 4.40 μm or less, more favorably 4.20 μm or less, still more favorably 4.00 μm or less, particularly favorably 3.80 μm or less, and most favorably 3.40 μm or less. When the upper limit value of the average thickness of the base **41** is 4.40 μm or less, it is possible to increase the recording capacity of one data cartridge to be more than that of a general magnetic tape. The lower limit value of the average thickness of the base **41** is favorably 3.00 μm or more, more favorably 3.20 μm or more. When the lower limit value of the average thickness of the base **41** is 3.00 μm or more, it is possible to suppress a decrease in the strength of the base **41**.

The average thickness of the base **41** is obtained as follows. First, the magnetic tape MT housed in the cartridge **10** is unwound, and the magnetic tape MT is cut into a length of 250 mm at a position 30 m to 40 m in the longitudinal direction from a connection part **21** between the magnetic tape MT and the leader tape LT to prepare a sample. In the present specification, the “longitudinal direction” in the “longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT” means the direction from one end on the side of the leader tape LT to the other end on the side opposite thereto.

Subsequently, the layers other than the base **41** of the sample (i.e., the underlayer **42**, the magnetic layer **43**, and

the back layer **44**) are removed with a solvent such as MEK (methyl ethyl ketone) and dilute hydrochloric acid. Next, the thickness of the sample (base **41**) is measured at five positions using a Laser Hologage (LGH-110C) manufactured by Mitutoyo Corporation as a measuring apparatus, and the measured values are simply averaged (arithmetically averaged) to calculate the average thickness of the base **41**. Note that the above five measurement positions are randomly selected from the sample such that they are different positions in the longitudinal direction of the magnetic tape MT.

The base **41** contains, for example, a polyester as a main component. The polyester include, for example, at least one selected from the group consisting of PET (polyethylene terephthalate), PEN (polyethylene naphthalate), PBT (polybutylene terephthalate), PBN (polybutylene naphthalate), PCT (polycyclohexylene dimethylene terephthalate), PEB (polyethylene-p(oxybenzoate), and polyethylene bisphenoxycarboxylate. In the case where the base **41** contains two or more types of polyesters, the two or more types of polyesters may be mixed, copolymerized, or stacked. At least one of an end or a side chain of the polyester may be modified. The base **41** may contain a resin described below other than polyesters, in addition to the polyester.

In the present specification, the “main component” means the component with the highest content ratio, of the components forming the base **41**. For example, in the case where the main component of the base **41** is a polyester, the content ratio of the polyester in the base **41** may be, for example, 50 mass % or more, 60 mass % or more, 70 mass % or more, 80 mass % or more, 90 mass % or more, 95 mass % or more, or 98 mass % or more with respect to the mass of the base **41**, or the base **41** may contain only the polyester.

The fact that the base **41** contains a polyester is confirmed, for example, as follows. First, as in the method of measuring the average thickness of the base **41**, the magnetic tape MT is prepared and cut into a length of 250 mm to prepare a sample, and then, the layers of the sample other than the base **41** are removed. Next, an IR spectrum of the sample (base **41**) is obtained by infrared absorption spectrometry (IR). On the basis of this IR spectrum, it can be confirmed that the base **41** contains a polyester.

The base **41** favorably contains a polyester. When the base **41** contains a polyester, the Young's modulus in the longitudinal direction of the base **41** can be reduced to favorably 2.5 GPa or more and 7.8 GPa or less, more favorably 3.0 GPa or more and 7.0 GPa or less. Therefore, by adjusting the tension in the longitudinal direction of the magnetic tape MT during travelling using the recording/reproduction apparatus, the width of the magnetic tape MT can be kept constant or substantially constant. The method of measuring the Young's modulus in the longitudinal direction of the base **41** will be described below.

The base **41** may contain a resin other than polyesters. In this case, the resin other than polyesters may be the main component of the material forming the base **41**. In the case where the resin other than polyesters is the main component of the material forming the base **41**, the content ratio of the resin other than polyesters in the base **41** may be, for example, 50 mass % or more, 60 mass % or more, 70 mass % or more, 80 mass % or more, 90 mass % or more, 95 mass % or more, or 98 mass % or more with respect to the mass of the base **41**, or the base **41** may contain only a resin other than polyesters. The resin other than polyesters includes, for example, at least one selected from the group consisting of a polyolefin resin, a cellulose derivative, a vinyl resin, and a different polymer resin. In the case where the base **41**

contains two or more of these resins, the two or more resins may be mixed, copolymerized, or stacked.

The polyolefin resin includes, for example, at least one selected from the group consisting of PE (polyethylene) and PP (polypropylene). The cellulose derivative includes, for example, at least one selected from the group consisting of cellulose diacetate, cellulose triacetate, CAB (cellulose acetate butyrate), and CAP (cellulose acetate propionate). The vinyl resin includes, for example, at least one selected from the group consisting of PVC (polyvinyl chloride) and PVDC (polyvinylidene chloride).

The different polymer resin includes, for example, at least one selected from the group consisting of PEEK (polyetheretherketone), PA (polyamide, nylon), aromatic PA (aromatic polyamide, aramid), PI (polyimide), aromatic PI (aromatic polyimide), PAI (polyamideimide), aromatic PAI (aromatic polyamideimide), PBO (polybenzoxazole, e.g., Zylon (registered trademark)), polyether, PEK (polyetherketone), polyetherester, PES (polyethersulfone), PEI (polyetherimide), PSF (polysulfone), PPS (polyphenylene sulfide), PC (polycarbonate), PAR (polyarylate), and PU (polyurethane). Specifically, the base **41** may contain, as a main component, PEEK (polyetheretherketone), PA (polyamide, nylon), aromatic PA (aromatic polyamide, aramid), PI (polyimide), aromatic PI (aromatic polyimide), PAI (polyamideimide), aromatic PAI (aromatic polyamideimide), PBO (polybenzoxazole, e.g., Zylon (registered trademark)), polyether, PEK (polyetherketone), polyetherester, PES (polyethersulfone), PEI (polyetherimide), PSF (polysulfone), PPS (polyphenylene sulfide), PC (polycarbonate), PAR (polyarylate), or PU (polyurethane).

The base **41** may be biaxially stretched in the longitudinal direction and the width direction. The polymer resin contained in the base **41** is favorably oriented in a direction oblique to the width direction of the base **41**. (Magnetic Layer)

The magnetic layer **43** is configured to be capable of recording signals by a magnetization pattern. The magnetic layer **43** may be a recording layer of a perpendicular recording type or a recording layer of a longitudinal recording type. The magnetic layer **43** includes magnetic particles, first particles, and second particles. The magnetic layer **43** may further include a binder. The magnetic layer **43** may further include at least one additive selected from the group consisting of a lubricant, a curing agent, a rust inhibitor, and non-magnetic reinforcing particles, as necessary.

Part B of FIG. 3 is a cross-sectional view showing an example of a configuration of a surface of the magnetic layer **43** (hereinafter, referred to as a "magnetic surface" in some cases.). The magnetic layer **43** has a large number of protrusions **430** on the magnetic surface. The large number of protrusions **430** include protrusions **430** formed by particles. Specifically, for example, the large number of protrusions **430** include protrusions **430** formed by first particles **431** and protrusions **430** formed by second particles **432**. The large number of particles forming the protrusions **430** may include at least one of particles covered with a binder or particles that are not covered with a binder and are partially exposed. Specifically, for example, the large number of first particles **431** forming the protrusions **430** may include at least one of the first particles **431** covered with a binder or the first particles **431** that are not covered with a binder and are partially exposed. The large number of second particles **432** forming the protrusions **430** may include at least one of the second particles **432** covered with a binder or the second particles **432** that are not covered with a binder and are partially exposed. Note that Part B of FIG.

3 shows an example in which the large number of first particles **431** forming the protrusions **430** are covered with a binder and the large number of second particles **432** forming the protrusions **430** are covered with a binder.

As shown in FIG. 4, the magnetic layer **43** may include a plurality of servo bands SB and a plurality of data bands DB in advance. The plurality of servo bands SB is provided at equal intervals in the width direction of the magnetic tape MT. The data band DB is provided between the servo bands SB adjacent to each other. The servo band SB is for guiding a head unit (magnetic head) **56** (specifically, servo read heads **56A** and **56B**) during recording or reproducing data. A servo pattern (servo signal) for tracking control of the head unit **56** is written to the servo band SB in advance. User data is recorded in the data band DB.

In order to read asymmetric servo stripes **113** (see FIG. 6) described below, the head unit **56** may be configured such that it can be kept inclined with respect to an axis Ax parallel to the width direction of the magnetic tape MT during recording and reproduction of data, as shown in FIG. 4. Alternatively, the head unit **56** may be configured to be inclined with respect to the above axis Ax in accordance with the meandering or deformation of the magnetic tape MT during recording and reproduction of data. The inclination angle of the head unit **56** with reference to the axis Ax parallel to the width direction of the magnetic tape MT is favorably 3° or more and 18° or less, more favorably 5° or more and 15° or less.

The upper limit value of a ratio R_S ($= (S_{SB}/S) \times 100$) of a total area S_{SB} of the plurality of servo bands SB to the area S of the magnetic surface (surface of the magnetic layer **43**) is favorably 4.0% or less, more favorably 3.5% or less, and still more favorably 3.0% or less, from the viewpoint of achieving high recording capacity. Meanwhile, the lower limit value of the ratio R_S of the total area S_{SB} of the plurality of servo bands SB to the area S of the magnetic surface is favorably 1.0% or more from the viewpoint of providing five or more servo bands SB.

The ratio R_S of the total area S_{SB} of the plurality of servo bands SB to the area S of the entire magnetic surface is obtained as follows. The magnetic tape MT is developed using a ferriccolloid developer (Sigmarker Q manufactured by Sigma Hi-Chemical Inc.), and then the developed magnetic tape MT is observed under an optical microscope to measure a servo band width W_{SB} and the number of servo bands SB. Next, the ratio R_S is obtained using the following formula.

$$\text{Ratio } R_S [\%] = \{ ((\text{servo band width } W_{SB}) \times (\text{number of servo bands SB})) / (\text{width of the magnetic tape MT}) \} \times 100$$

The number of servo bands SB is, for example, $5+4n$ (where n represents an integer equal to or greater than 0.). The number of servo bands SB is favorably 5 or more, more favorably 9 or more. When the number of servo bands SB is 5 or more, it is possible to prevent the dimensional change of in the width direction of the magnetic tape MT from affecting the servo signal, and achieve stable recording and reproduction properties with less off-track. The upper limit value of the number of servo bands SB is not particularly limited, but is, for example, 33 or less.

The number of servo bands SB is obtained in the same manner as the above method of calculating the ratio R_S .

The upper limit value of the servo band width W_{SB} is favorably 95 μm or less, more favorably 65 μm or less, and still more favorably 50 μm or less from the viewpoint of achieving high recording capacity. The lower limit value of the servo band width W_{SB} is favorably 10 μm or more. It is difficult to produce a magnetic head that is capable of reading a servo signal with the servo band width W_{SB} of less than 10 μm .

The width of the servo band width W_{SB} is obtained in the same manner as the above method of calculating the ratio R_S .

As shown in FIG. 5, the magnetic layer 43 is configured such that a plurality of data tracks Tk can be formed in the data band DB. The upper limit value of a data track width W is favorably 1200 nm or less, more favorably 1000 nm or less, still more favorably 800 nm or less, and particularly favorably 600 nm or less, from the viewpoint of improving track recording density and achieving high recording capacity. The lower limit value of the data track width W is favorably 20 nm or more considering the size of magnetic particles.

The magnetic layer 43 is configured to be capable of recording data such that the minimum value L of a distance between magnetization reversals is favorably 47 nm or less, more favorably 45 nm or less, and still more favorably 40 nm or less, from the viewpoint of achieving high recording capacity. The lower limit value of the minimum value L of the distance between magnetization reversals is favorably 20 nm or more considering the size of magnetic particles.

The data track width W is obtained as follows. First, the cartridge 10 with data recorded on the entire surface of the magnetic tape MT is prepared, the magnetic tape MT is unwound from this cartridge 10, and the magnetic tape MT is cut into a length of 250 mm at a position 30 m to 40 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT to prepare a sample. Subsequently, the data recording pattern of the data band DB portion of the magnetic layer 43 of the sample is observed using a magnetic force microscope (MFM) to obtain an MFM image. Dimension 3100 manufactured by Digital Instruments and the analysis software thereof are used as the MFM. The measurement region of the MFM image is 10 $\mu\text{m} \times 10 \mu\text{m}$, and the measurement region of 10 $\mu\text{m} \times 10 \mu\text{m}$ is divided into 512 $\times$ 512 (=262,144) measurement points. The measurement by MFM is performed on three 10 $\mu\text{m} \times 10 \mu\text{m}$ measurement regions with different locations, i.e. three MFM images are obtained. From the three obtained MFM images, the track widths are measured at 10 locations using the analysis software attached to Dimension 3100, and the average value thereof (which is a simple average) is obtained. The obtained average value is the data track width W. Note that the measurement conditions of the above MFM are as follows: sweep rate: 1 Hz, chip used: MFMR-20, lift height: 20 nm, and correction: Flatten order 3.

The minimum value L of the distance between magnetization reversals is obtained as follows. First, the cartridge 10 with data recorded on the entire surface of the magnetic tape MT is prepared, the magnetic tape MT is unwound from this cartridge 10, and the magnetic tape MT is cut into a length of 250 mm at a position 30 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT to prepare a sample. Subsequently, the data recording pattern of the data band DB portion of the magnetic layer 43 of the sample is observed using a magnetic force microscope (MFM) to obtain an MFM image. Dimension 3100 manufactured by Digital Instruments and

the analysis software thereof are used as the MFM. The measurement region of the MFM image is 2 $\mu\text{m} \times 2 \mu\text{m}$, and the measurement region of 2 $\mu\text{m} \times 2 \mu\text{m}$ is divided into 512 $\times$ 512 (=262,144) measurement points. The measurement by MFM is performed on three 2 $\mu\text{m} \times 2 \mu\text{m}$ measurement regions with different locations, i.e. three MFM images are obtained. Fifty bit-to-bit distances are measured from the two-dimensional uneven chart of the recording pattern of the obtained MFM image. The bit-to-bit distances are measured using the analysis software attached to Dimension 3100. The value that becomes approximately the greatest common divisor of the measured 50 bit-to-bit distances is defined as the minimum value L of the distance between magnetization reversals. Note that the measurement conditions are as follows: sweep rate: 1 Hz, chip used: MFMR-20, lift height: 20 nm, and correction: Flatten order 3.

The servo pattern is a magnetized region and is formed by magnetizing a specific region of the magnetic layer 43 in a specific direction by a servo write head during production of a magnetic tape. Of the servo band SB, a region in which the servo pattern is not formed (hereinafter, referred to as a "non-pattern region".) may be a magnetized region in which the magnetic layer 43 is magnetized or a non-magnetized region in which the magnetic layer 43 is not magnetized. In the case where the non-pattern region is a magnetized region, the servo pattern forming region and the non-pattern region are magnetized in different directions (e.g., opposite directions).

In the LTO standard, as shown in FIG. 6, a servo pattern that includes a plurality of servo stripes (linear magnetized regions) 113 inclined with respect to the axis Ax parallel to the width direction of the magnetic tape MT is formed in the servo band SB.

The servo band SB includes a plurality of servo frames 110. Each servo frame 110 includes 18 servo stripes 113. Specifically, each servo frame 110 includes a servo subframe 1 (111) and a servo subframe 2 (112).

The servo subframe 1 (111) includes an A burst 111A and a B burst 111B. The B burst 111B is disposed adjacent to the A burst 111A. The A burst 111A includes five servo stripes 113 formed at predetermined intervals and inclined at a predetermined angle θ_1 with respect to the axis Ax parallel to the width direction of the magnetic tape MT. In FIG. 6, these five servo stripes 113 are indicated by reference symbols A₁, A₂, A₃, A₄, and A₅ from the EOT (End Of Tape) to the BOT (Beginning Of Tape) of the magnetic tape MT.

The B burst 111B includes five servo stripes 113 formed at predetermined intervals and inclined at a predetermined angle θ_2 with respect to the axis Ax parallel to the width direction of the magnetic tape MT. In FIG. 6, these five servo stripes 113 are denoted by reference symbols B₁, B₂, B₃, B₄, and B₅ from the EOT to the BOT of the magnetic tape MT.

The servo stripes 113 of the B burst 111B are inclined in the opposite direction to the servo stripes 113 of the A burst 111A. The servo stripes 113 of the A burst 111A and the servo stripes 113 of the B burst 111B are asymmetric with respect to the axis Ax parallel to the width direction of the magnetic tape MT. That is, the servo stripes 113 of the A burst 111A and the servo stripes 113 of the B burst 111B are disposed in a substantially inverted V-shape. Since the servo stripes 113 of the A burst 111A and the servo stripes 113 of the B burst 111B are asymmetric with respect to the axis Ax, there is a state in which the servo stripes 113 of the A burst 111A and the servo stripes 113 of the B burst 111B are substantially symmetric with respect to the central axis of the sliding surface of the head unit 56 when the head unit 56 is inclined with respect to the axis Ax. By changing the

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inclination of the head unit **56** with reference to this state, it is possible to adjust the distance between the servo read heads **56A** and **56B** in the width direction of the magnetic tape MT. Therefore, in both cases where the width of the magnetic tape MT is widened and where the width of the magnetic tape MT is narrowed, it is possible to cause the servo read heads **56A** and **56B** to face predetermines position of the servo band SB. Note that the central axis of the sliding surface of the head unit **56** means an axis that passes through the center of the plurality of servo read heads **56A** and **56B** on the sliding surface of the head unit **56**.

The predetermined angle θ_1 that is the inclination angle of the servo stripes **113** of the A burst **111A** and the predetermined angle θ_2 that is the inclination angle of the servo stripes **113** of the B burst **111B** are different from each other. More specifically, the predetermined angle θ_1 of the servo stripes **113** of the A burst **111A** may be larger than the predetermined angle θ_2 of the servo stripes **113** of the B burst **111B**, or the predetermined angle θ_2 of the servo stripes **113** of the B burst **111B** may be larger than the predetermined angle θ_1 of the servo stripes **113** of the A burst **111A**. That is, the inclination of the servo stripes **113** of the A burst **111A** may be larger than the inclination of the servo stripes **113** of the B burst **111B**, or the inclination of the servo stripes **113** of the B burst **111B** may be larger than the inclination of the servo stripes **113** of the A burst **111A**. Note that FIG. 6 shows an example in which the predetermined angle θ_1 of the servo stripes **113** of the A burst **111A** is larger than the predetermined angle θ_2 of the servo stripes **113** of the B burst **111B**. A case where the predetermined angle θ_1 of the servo stripes **113** of the A burst **111A** is larger than the predetermined angle θ_2 of the servo stripes **113** of the B burst **111B** will be described below.

The servo subframe 2 (**112**) includes a C burst **112C** and a D burst **112D**. The D burst **112D** is disposed adjacent to the C burst **112C**. The C burst **112C** includes four servo stripes **113** formed at predetermined intervals and inclined at the predetermined angle θ_1 with respect to the axis Ax parallel to the width direction of the magnetic tape MT. In FIG. 6, these four servo stripes **113** are denoted by reference symbols C_1 , C_2 , C_3 , and C_4 from the EOT to the BOT of the magnetic tape MT.

The D burst **112D** includes four servo stripes **113** formed at predetermined intervals and inclined at the predetermined angle θ_2 with respect to the axis Ax parallel to the width direction of the magnetic tape MT. In FIG. 6, these four servo stripes **113** are denoted by D_1 , D_2 , D_3 , and D_4 from the EOT to the BOT of the magnetic tape MT.

The servo stripes **113** of the D burst **112D** are inclined in the opposite direction to the servo stripes **113** of the C burst **112C**. The servo stripes **113** of the C burst **112C** and the servo stripes **113** of the D burst **112D** are asymmetric with respect to the axis Ax parallel to the width direction of the magnetic tape MT. That is, the servo stripes **113** of the C burst **112C** and the servo stripes **113** of the D burst **112D** are disposed in a substantially inverted V-shape. Since the servo stripes **113** of the C burst **112C** and the servo stripes **113** of the D burst **112D** are substantially symmetric with respect to the central axis of the head unit **56** when the head unit **56** is inclined with respect to the axis Ax. By changing the inclination of the head unit **56** with reference to this state, it is possible to adjust the distance between servos.

The predetermined angle θ_1 that is the inclination angle of the servo stripes **113** of the C burst **112C** and the predeter-

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mined angle θ_2 that is the inclination angle of the servo stripes **113** of the D burst **112D** are different from each other. More specifically, the predetermined angle θ_1 of the servo stripes **113** of the C burst **112C** may be larger than the predetermined angle θ_2 of the servo stripes **113** of the D burst **112D**, or the predetermined angle θ_2 of the servo stripes **113** of the D burst **112D** may be larger than the predetermined angle θ_1 of the servo stripes **113** of the C burst **112C**. That is, the inclination of the servo stripes **113** of the C burst **112C** may be larger than the inclination of the servo stripes **113** of the D burst **112D**, or the inclination of the servo stripes **113** of the D burst **112D** may be larger than the inclination of the servo stripes **113** of the C burst **112C**. Note that FIG. 6 shows an example in which the predetermined angle θ_1 of the servo stripes **113** of the C burst **112C** is larger than the predetermined angle θ_2 of the servo stripes **113** of the D burst **112D**. A case where the predetermined angle θ_1 of the servo stripes **113** of the C burst **112C** is larger than the predetermined angle θ_2 of the servo stripes **113** of the D burst **112D** will be described below.

The above predetermined angle θ_1 of the servo stripes **113** in the A burst **111A** and the C burst **112C** is favorably 180 or more and 280 or less, more favorably 180 or more and 260 or less. The above predetermined angle θ_2 of the servo stripes **113** in the B burst **111B** and the D burst **112D** is favorably -4° or more and 6° or less, more favorably -2° or more and 6° or less. The servo stripes **113** in the A burst **111A** and the C burst **112C** are an example of the first magnetized region. The servo stripe **113** in the B burst **111B** and the D burst **112D** are an example of the second magnetized region.

By reading the servo band SB with the head unit **56**, information for obtaining the tape speed and the position of the head unit **56** in the longitudinal direction is acquired. The tape speed is calculated from the time between four timing signals ($A1-C1$, $A2-C2$, $A3-C3$, and $A4-C4$). The head position is calculated from the above-mentioned time between four timing signals and the time between other four timing signals ($A1-B1$, $A2-B2$, $A3-B3$, and $A4-B4$). The servo pattern may have a shape including two parallel lines.

As shown in FIG. 6, the servo pattern (i.e., the plurality of servo stripes **113**) is favorably arranged linearly in the longitudinal direction of the magnetic tape MT. That is, the servo band SB is favorably linear in the longitudinal direction of the magnetic tape MT.

The upper limit value of an average thickness t_i of the magnetic layer **43** is favorably 80 nm or less, more favorably 65 nm or less, and still more favorably 55 nm or less. When the upper limit value of the average thickness t_i of the magnetic layer **43** is 80 nm or less, the effect of the demagnetizing field can be reduced in the case of using a ring-type head as a recording head, and thus, further excellent electromagnetic conversion characteristics can be achieved.

The lower limit value of the average thickness t_l of the magnetic layer **43** is favorably 35 nm or more. When the lower limit value of the average thickness t_l of the magnetic layer **43** is 35 nm or more, the output can be secured in the case of using an MR-type head as a reproduction head, and thus, further excellent electromagnetic conversion characteristics can be achieved.

The average thickness t_l of the magnetic layer **43** is obtained as follows. First, the magnetic tape MT housed in the cartridge **10** is unwound, and the magnetic tape MT is cut into a length of 250 mm at positions in the longitudinal direction 10 m to 20 m, 30 m to 40 m, and 50 m to 60 m from the connection part **21** between the magnetic tape MT and

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the leader tape LT to prepare three samples. Subsequently, each sample is processed by an FIB method or the like for slicing. In the case of using an FIB method, a carbon layer and a tungsten layer are formed as protective layers as pre-processing for observing a TEM image of a cross section described below. The carbon layer is formed on the surfaces of the magnetic tape MT on the side of the magnetic layer **43** and on the side of the back layer **44** by a vapor deposition method, and the tungsten layer is further formed on the surface on the side of the magnetic layer **43** by a vapor deposition method or a sputtering method. The slicing is performed along the longitudinal direction of the magnetic tape MT. That is, the slicing forms a cross section parallel to both the longitudinal direction and thickness direction of the magnetic tape MT.

The above cross section of each obtained sliced sample is observed under the following conditions using a transmission electron microscope (TEM) to obtain a TEM image of each sliced sample. Note that the magnification and the acceleration voltage may be adjusted as appropriately in accordance with the type of apparatus.

Apparatus: TEM (H9000NAR manufactured by Hitachi, Ltd.)

Acceleration voltage: 300 kV

Magnification: 100,000 times

Next, the obtained TEM image of each sliced sample is used to measure the thickness of the magnetic layer **43** at 10 positions of each sliced sample. Note that the 10 measurement positions of each sliced sample are randomly selected from the sample such that they are different positions in the longitudinal direction of the magnetic tape MT. An average value obtained by simply averaging (arithmetically averaging) the measured values (30 thicknesses of the magnetic layer **43** in total) of each obtained sliced sample is used as the average thickness t_1 [nm] of the magnetic layer **43**. (Magnetic Particles)

The magnetic particles are, for example, particles containing hexagonal ferrite (hereinafter, referred to as "hexagonal ferrite particles"), particles containing epsilon-type iron oxide (ϵ -iron oxide) (hereinafter, referred to as " ϵ -iron oxide particles"), or particles containing Co-containing spinel ferrite (hereinafter, referred to as "cobalt ferrite particles"). It is favorable that the magnetic particles are preferentially crystal-oriented in the perpendicular direction of the magnetic tape MT. In the present specification, the perpendicular direction (thickness direction) of the magnetic tape MT means the thickness direction of the magnetic tape MT in a flat state.

(Hexagonal Ferrite Particles)

The hexagonal ferrite particles each have, for example, a plate shape such as a hexagonal plate shape or a columnar shape such as a hexagonal columnar shape (however, the thickness or height is smaller than the long diameter of the plate surface or bottom surface.). In the present specification, the hexagonal plate shape includes a substantially hexagonal plate shape. The hexagonal ferrite favorably includes at least one selected from the group consisting of Ba, Sr, Pb, and Ca, more favorably at least one selected from the group consisting of Ba and Sr. Specifically, the hexagonal ferrite may be, for example, barium ferrite or strontium ferrite. The barium ferrite may further include at least one selected from the group consisting of Sr, Pb, and Ca, in addition to Ba. The strontium ferrite may further include at least one selected from the group consisting of Ba, Pb, and Ca, in addition to Sr.

More specifically the hexagonal ferrite has an average composition represented by a general formula $MFe_{12}O_{19}$.

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However, M represents, for example, at least one metal selected from the group consisting of Ba, Sr, Pb, and Ca, favorably at least one metal selected from the group consisting of Ba and Sr. M may represent a combination of Ba and at least one metal selected from the group consisting of Sr, Pb, and Ca. Further, M may represent a combination of Sr and at least one metal selected from the group consisting of Ba, Pb, and Ca. In the above general formula, some Fes may be substituted with another metal element.

In the case where the magnetic particles include hexagonal ferrite particles, the average particle size of the magnetic particles is favorably 13 nm or more and 22 nm or less, more favorably 13 nm or more and 19 nm or less, still more favorably 13 nm or more and 18 nm or less, particularly favorably 14 nm or more and 17 nm or less, and most favorably 14 nm or more and 16 nm or less. When the average particle size of the magnetic particles is 22 nm or less, it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR) in the magnetic tape MT having high recording density. Meanwhile, when the average particle size of the magnetic particles is 13 nm or more, the dispersibility of the magnetic particles is improved, and it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR).

In the case where the magnetic particles include hexagonal ferrite particles, the average aspect ratio of the magnetic particles is favorably 1.0 or more and 3.0 or less, more favorably 1.5 or more and 2.8 or less, and still more favorably 1.8 or more and 2.7 or less. When the average aspect ratio of the magnetic particles is within the range of 1.0 or more and 3.0 or less, the aggregation of the magnetic particles can be suppressed. Further, when the magnetic particles are perpendicularly oriented in the process of forming the magnetic layer **43**, the resistance to be applied to the magnetic particles can be suppressed. Therefore, it is possible to improve the perpendicular orientation of the magnetic particles.

In the case where the magnetic particles include hexagonal ferrite particles, the average particle size and the average aspect ratio of the magnetic particles are obtained as follows. First, the magnetic tape MT housed in the cartridge **10** is unwound, and the magnetic tape MT is cut out at a position 30 m to 40 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT. Subsequently, the magnetic tape MT to be measured is processed by an FIB method or the like for slicing. In the case of using an FIB method, a carbon layer and a tungsten layer are formed as protective layers as pre-processing for observing a TEM image of a cross section described below. The carbon layer is formed on the surfaces of the magnetic tape MT on the side of the magnetic layer **43** and on the side of the back layer **44** by a vapor deposition method, and the tungsten layer is further formed on the surface on the side of the magnetic layer **43** by a vapor deposition method or a sputtering method. The slicing is performed along the length direction (longitudinal direction) of the magnetic tape MT. That is, the slicing forms a cross section parallel to both the longitudinal direction and thickness direction of the magnetic tape MT.

The above cross section of the obtained sliced sample is observed using a transmission electron microscope (H-9500 manufactured by Hitachi High-Technologies Corporation) at an acceleration voltage: 200 kV and a total magnification of 500,000 times such that the entire magnetic layer **43** is included in the thickness direction of the magnetic layer **43** to take a TEM image. TEM images are prepared in such a

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number that 50 particles for which a plate diameter DB and a plate thickness DA (see FIG. 7) shown below can be measured can be extracted.

In the present specification, regarding the size of the hexagonal ferrite particles (hereinafter, referred to as "particle size"), in the case where the shape of each particle observed in the above TEM image is a plate shape or a columnar shape (however, the thickness or height is smaller than the long diameter of the plate surface or bottom surface.) as shown in FIG. 7, the long diameter of the plate surface or bottom surface is used as the value of the plate diameter DB. The thickness or height of the particle observed in the above TEM image is used as the value of the plate thickness DA. In the case where the plate surface or bottom surface of the particle observed in the TEM image has a hexagonal shape, the long diameter means a diagonal distance. In the case where the thickness or height of one particle is not constant in the particle, the maximum thickness or height of the particle is used as the plate thickness DA.

Next, 50 particles to be extracted from the taken TEM image are selected on the basis of the following criteria. Particles that are partially outside the field of view of the TEM image are not measured, and particles that have a clear outline and are present in isolation are measured. In the case where particle overlap with each other, if the boundary between the particles is clear and the shape of the entire particle can be determined, each particle is measured as a single particle. However, particles whose boundaries are unclear and whose overall shapes are also unclear are not measured as the shape of the particle cannot be determined.

FIG. 8 and FIG. 9 respectively show a first example and a second example of the TEM image. In FIG. 8 and FIG. 9, for example, particles indicated by arrows a and d are selected because their plate thickness (thickness or height of the particle) DA can be clearly checked. The plate thickness DA of each of the 50 selected particles is measured. The plate thicknesses DA thus obtained are simply averaged (arithmetically averaged) to obtain an average plate thickness DA_{ave} . The average plate thickness DA_{ave} is an average particle plate thickness. Subsequently, the plate diameter DB of each magnetic particle is measured. In order to measure the plate diameter DB of the particle, 50 particles whose plate diameter DB can be clearly checked are selected from the taken TEM image. For example, in FIG. 8 and FIG. 9, for example, particles indicated by arrows b and c are selected because the plate diameter DB can be clearly checked. The plate diameter DB of each of the 50 selected particles is measured. The plate diameters DB obtained in this way are simply averaged (arithmetically averaged) to obtain an average plate diameter DB_{ave} . The average plate diameter DB_{ave} is the average particle size. Then, the average aspect ratio (DB_{ave}/DA_{ave}) of the particles is obtained on the basis of the average plate thickness DA_{ave} and the average plate diameter DB_{ave} .

In the case where the magnetic particles include hexagonal ferrite particles, the average particle volume of the magnetic particles is favorably 500 nm³ or more and 2500 nm³ or less, more favorably 500 nm³ or more and 1600 nm³ or less, still more favorably 500 nm³ or more and 1500 nm³ or less, particularly favorably 600 nm³ or more and 1200 nm³ or less, and most favorably 600 nm³ or more and 1000 nm³ or less. When the average particle volume of the magnetic particles is 2500 nm³ or less, effects similar to those in the case where the average particle size of the magnetic particles is 22 nm or less can be achieved. Meanwhile, when the average particle volume of the magnetic

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particles is 500 nm³ or more, effects similar to those in the case where the average particle size of the magnetic particles is 13 nm or more can be achieved.

The average particle volume of the magnetic particles is obtained as follows. First, the average plate thickness DA_{ave} and the average plate diameter DB_{ave} are obtained as described above regarding the method of calculating the average particle size of the magnetic particles. Next, an average volume V of the magnetic particles is obtained by the following formula.

$$V = \frac{3\sqrt{3}}{8} \times DA_{ave} \times DB_{ave} \times DB_{ave} \quad (\text{Math. 1})$$

(ε-Iron Oxide Particles)

The ε-iron oxide particles are hard magnetic particles capable of achieving a high coercive force even as fine particles. The ε-iron oxide particles each have a spherical shape or a cubic shape. In the present specification, the spherical shape includes a substantially spherical shape. Further, the cubic shape includes a substantially cubic shape. Since the ε-iron oxide particles each have the above shape, in the case where ε-iron oxide particles are used as magnetic particles, the contact area between particles in the thickness direction of the magnetic tape MT can be reduced and the aggregation of the particles can be suppressed as compared with the case where barium ferrite particles each having a hexagonal plate shape are used as magnetic particles. Therefore, it is possible to improve the dispersibility of the magnetic particles and achieve further excellent electromagnetic conversion characteristics (e.g., SNR).

The ε-iron oxide particles may each have a structure of a composite particle. More specifically, the ε-iron oxide particles each include an ε-iron oxide portion and a portion having soft magnetism or a magnetic portion having a saturation magnetization amount σ_s higher than that of ε-iron oxide and a coercive force H_c smaller than that of ε-iron oxide (hereinafter, referred to as a "portion having soft magnetism or the like").

The ε-iron oxide portion contains ε-iron oxide. The ε-iron oxide contained in the ε-iron oxide portion favorably has ε-Fe₂O₃ crystals as the main phase, more favorably single-phase ε-Fe₂O₃.

The portion having soft magnetism or the like is at least partially in contact with the ε-iron oxide portion. Specifically, the portion having soft magnetism or the like may partially cover the periphery of the ε-iron oxide portion or may cover the entire periphery of the ε-iron oxide portion.

The portion having soft magnetism contains a soft magnetic material such as α-Fe, a Ni—Fe alloy, and a Fe—Si—Al alloy. α-Fe may be one obtained by reducing ε-iron oxide contained in the ε-iron oxide portion.

The magnetic portion having a saturation magnetization amount σ_s higher than that of ε-iron oxide and a coercive force H_c smaller than that of ε-iron oxide may contain, for example, Fe₃O₄, γ-Fe₂O₃, or spinel ferrite.

When the ε-iron oxide particle includes the portion having soft magnetism or the like as described above, the coercive force H_c of the entire ε-iron oxide particle (composite particle) can be adjusted to a coercive force H_c suitable for recording while maintaining the coercive force H_c of the ε-iron oxide portion alone at a large value in order to ensure thermal stability.

The ε-iron oxide particles may include an additive instead of the above structure of the composite particle or may

include an additive while having the above structure of the composite particle. In this case, some Fe of the ϵ -iron oxide particles are substituted with the additives. Also with the ϵ -iron oxide particles including the additives, the coercive force H_c of the entire ϵ -iron oxide particles can be adjusted to a coercive force H_c suitable for recording, and thus, it is possible to improve the easiness of recording. The additive is a metal element other than iron, favorably a trivalent metal element, more favorably at least one selected from the group consisting of Al, Ga, and In, and still more favorably at least one selected from the group consisting of Al and Ga.

Specifically, the ϵ -iron oxide including the additive is an ϵ - $\text{Fe}_{2-x}\text{M}_x\text{O}_3$ crystal (where M represents a metal element other than iron, favorably a trivalent metal element, more favorably more favorably at least one selected from the group consisting of Al, Ga, and In, and still more favorably at least one selected from the group consisting of Al and Ga. x satisfies the following formula represented by: $0 < x < 1$, for example.).

In the case where the magnetic particles include ϵ -iron oxide particles, the average particle size of the magnetic particles is favorably 10 nm or more and 20 nm or less, more favorably 10 nm or more and 18 nm or less, still more favorably 10 nm or more and 16 nm or less, particularly favorably 10 nm or more and 15 nm or less, and most favorably 10 nm or more and 14 nm or less. In the magnetic tape MT, the region half the size of the recording wavelength is an actual magnetized region. For this reason, by setting the average particle size of the magnetic particles to half or less of the shortest recording wavelength, it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR). Therefore, when the average particle size of the magnetic particles is 20 nm or less, it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR) in the magnetic tape MT with high recording capacity (e.g., the magnetic tape MT configured to be capable of recording signals at the shortest recording wavelength of 40 nm or less). Meanwhile, when the average particle size of the magnetic particles is 10 nm or more, the dispersibility of the magnetic particles is improved, and it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR).

In the case where the magnetic particles include ϵ -iron oxide particles, the average aspect ratio of the magnetic particles is favorably 1.0 or more and 3.0 or less, more favorably 1.0 or more and 2.5 or less, still more favorably 1.0 or more and 2.1 or less, and particularly favorably 1.0 or more and 1.8 or less. When the average aspect ratio of the magnetic particles is within the range of 1.0 or more and 3.0 or less, the aggregation of the magnetic particles can be suppressed. Further, when the magnetic particles are perpendicularly oriented in the process of forming the magnetic layer 43, the resistance to be applied to the magnetic particles can be suppressed. Therefore, it is possible to improve the perpendicular orientation of the magnetic particles.

In the case where the magnetic particles include ϵ -iron oxide particles, the average particle size and the average aspect ratio of the magnetic particles are obtained as follows. First, the magnetic tape MT housed in the cartridge 10 is unwound, and the magnetic tape MT is cut out at a position 30 m to 40 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT. Subsequently, the magnetic tape MT to be measured is processed by an FIB (Focused Ion Beam) method or the like for slicing. In the case of using an FIB method, a carbon layer and a tungsten layer are formed as protective layers as

pre-processing for observing a TEM image of a cross section described below. The carbon layer is formed on the surfaces of the magnetic tape MT on the side of the magnetic layer 43 and on the side of the back layer 44 by a vapor deposition method, and the tungsten layer is further formed on the surface on the side of the magnetic layer 43 by a vapor deposition method or a sputtering method. The slicing is performed along the length direction (longitudinal direction) of the magnetic tape MT. That is, the slicing forms a cross section parallel to both the longitudinal direction and thickness direction of the magnetic tape MT.

The above cross section of the obtained sliced sample is observed using H-9500 manufactured by Hitachi High-Technologies Corporation at an acceleration voltage: 200 kV and a total magnification of 500,000 times such that the entire magnetic layer 43 is included in the thickness direction of the magnetic layer 43 to take a TEM image. Next, 50 particles whose shapes can be clearly checked are selected from the take TEM image and a major axis length DL and a minor axis length DS of each particle are measured. Here, the major axis length DL means the largest one (so-called maximum Feret diameter) of the distances between two parallel lines drawn from all angles so as to be in contact with the contour of each particle. Meanwhile, the minor axis length DS means the largest one of the lengths of the particles in the direction perpendicular to the major axis (DL) of the particles. Subsequently, the measured major axis lengths DL of the 50 particles are simply averaged (arithmetically averaged) to obtain an average major axis length DL_{ave} . The average major axis length DL_{ave} obtained in this way is the average particle size of the magnetic particles. Further, the measured minor axis lengths DS of the 50 particles are simply averaged (arithmetically averaged) to obtain an average minor axis length DS_{ave} . Then, an average aspect ratio (DL_{ave}/DS_{ave}) of the particles is obtained on the basis of the average major axis length DL_{ave} and the average short-axis length DS_{ave} .

In the case where the magnetic particles include ϵ -iron oxide particles, the average particle volume of the magnetic particles is favorably 500 nm³ or more and 4000 nm³ or less, more favorably 500 nm³ or more and 3000 nm³ or less, still more favorably 500 nm³ or more and 2000 nm³ or less, particularly favorably 600 nm³ or more and 1600 nm³ or less, and most favorably 600 nm³ or more and 1300 nm³ or less. In general, since the noise of the magnetic tape MT is inversely proportional to the square root of the number of particles (i.e., proportional to the square root of the particle volume), further excellent electromagnetic conversion characteristics (e.g., SNR) can be achieved by making the particle volume smaller. Therefore, when the average particle volume of the magnetic particles is 4000 nm³ or less, further excellent electromagnetic conversion characteristics (e.g., SNR) can be achieved similarly to the case where the average particle size of the magnetic particles is 20 nm or less. Meanwhile, when the average particle volume of the magnetic particles is 500 nm³ or more, effects similar to those in the case where the average particle size of the magnetic particles is 10 nm or more can be achieved.

In the case where ϵ -iron oxide particles each have a spherical shape, the average particle volume of the magnetic particles is obtained as follows. First, the average major axis length DL_{ave} in the same manner as the above method of calculating the average particle size of the magnetic particles. Next, an average volume V of the magnetic particles is obtained by the following formula.

$$V = (\pi/6) \times DL_{ave}^3$$

In the case where the ϵ -iron oxide particles each have a cubic shape, the average particle volume of the magnetic particles is obtained as follows. First, the magnetic tape MT housed in the cartridge 10 is unwound, and the magnetic tape MT is cut out at a position 30 m to 40 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT. Subsequently, the cut magnetic tape MT is processed by an FIB (Focused Ion Beam) or the like for slicing. In the case of using an FIB method, a carbon film and a tungsten thin film are formed as protective films as pre-processing for observing a TEM image of a cross section described below. The carbon film is formed on the surfaces of the magnetic tape MT on the side of the magnetic layer 43 and on the side of the back layer 44 by a vapor deposition method, and the tungsten thin film is further formed on the surface on the side of the magnetic layer 43 by a vapor deposition method or a sputtering method. The slicing is performed along the length direction (longitudinal direction) of the magnetic tape MT. That is, the slicing forms a cross section parallel to both the longitudinal direction and thickness direction of the magnetic tape MT.

The cross section of the obtained sliced sample is observed using H-9500 manufactured by Hitachi High-Technologies Corporation at an acceleration voltage: 200 kV and a total magnification of 500,000 times such that the entire magnetic layer 43 is included in the thickness direction of the magnetic layer 43 to obtain a TEM image. Note that the magnification and the acceleration voltage may be adjusted as appropriately in accordance with the type of apparatus. Next, 50 particles whose shapes are clear are selected from the taken TEM image and a side length DC of each particle is measured. Subsequently, the measured side lengths DC of the 50 particles are simply averaged (arithmetically averaged) to obtain an average side length DC_{ave} . Next, an average volume V_{ave} (particle volume) of the magnetic particles is obtained by the following formula using the average side length DC_{ave} .

$$V_{ave} = DC_{ave}^3$$

(Cobalt Ferrite Particles)

It is favorable that the cobalt ferrite particles each have uniaxial crystal anisotropy. When the cobalt ferrite particle has uniaxial crystal anisotropy, the magnetic particles can be preferentially crystal-oriented in the perpendicular direction of the magnetic tape MT. The cobalt ferrite particle has, for example, a cubic shape. In the present specification, the cubic shape includes a substantially cubic shape. The Co-containing spinel ferrite may further include at least one selected from the group consisting of Ni, Mn, Al, Cu, and Zn, in addition to Co.

The Co-containing spinel ferrite has an average composition represented by the following formula, for example.



(in the formula, M represents, for example, at least one metal selected from the group consisting of Ni, Mn, Al, Cu, and Zn. x represents a value within the range of $0.4 \leq x \leq 1.0$. y represents a value within the range of $0 \leq y \leq 0.3$. However, x and y satisfy the relationship represented by the following formula: $(x+y) \leq 1.0$. z represents a value within the range of $3 \leq z \leq 4$. Some Fes may be substituted with other metal elements.)

In the case where the magnetic particles include cobalt ferrite particles, the average particle size of the magnetic particles is favorably 8 nm or more and 16 nm or less, more favorably 8 nm or more and 13 nm or less, and still more favorably 8 nm or more and 10 nm or less. When the average

particle size of the magnetic particles is 16 nm or less, it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR) in the magnetic tape MT having high recording density. Meanwhile, when the average particle size of the magnetic particles is 8 nm or more, the dispersibility of the magnetic particles is improved, and it is possible to achieve further excellent electromagnetic conversion characteristics (e.g., SNR). The method of calculating the average particle size of the magnetic particles is similar to the method of calculating the average particle size of the magnetic particles in the case where the magnetic particles include ϵ -iron oxide particles.

In the case where the magnetic particles include cobalt ferrite particles, the average aspect ratio of the magnetic particles is favorably 1.0 or more and 3.0 or less, more favorably 1.0 or more and 2.5 or less, and still more favorably 1.0 or more and 2.0 or less. When the average aspect ratio of the magnetic particles is within the range of 1.0 or more and 3.0 or less, the aggregation of the magnetic particles can be suppressed. Further, when the magnetic particles are perpendicularly oriented in the process of forming the magnetic layer 43, the resistance to be applied to the magnetic particles can be suppressed. Therefore, it is possible to improve the perpendicular orientation of the magnetic particles. The method of calculating the average aspect ratio of the magnetic particles is similar to the method of calculating the average aspect ratio of the magnetic particles in the case where the magnetic particles include a powder of ϵ -iron oxide particles.

In the case where the magnetic particles include a powder of cobalt ferrite particles, the average particle volume of the magnetic particles is favorably 500 nm^3 or more and 4000 nm^3 or less, more favorably 600 nm^3 or more and 2000 nm^3 or less, and still more favorably 600 nm^3 or more and 1000 nm^3 or less. When the average particle volume of the magnetic particles is 4000 nm^3 or less, effects similar to those in the case where the average particle size of the magnetic particles is 16 nm or less can be achieved. Meanwhile, when the average particle volume of the magnetic particles is 500 nm^3 or more, effects similar to those in the case where the average particle size of the magnetic particles is 8 nm or more can be achieved. The method of calculating the average particle volume of the magnetic powder is similar to the method of calculating the average particle volume in the case where the ϵ -iron oxide particles each have a cubic shape.

(First Particle)

As shown in Part B of FIG. 3, some of the first particles 431 included in the magnetic layer 43 protrude from the magnetic surface to form the protrusion 430. When the head unit 56 slides over the magnetic tape MT, the protrusion 430 formed by the first particle 431 is capable of coming into contact with the head unit 56.

The first particle 431 has conductivity. The first particle 431 is favorably an antistatic agent and a solid lubricant. The first particle 431 is a particle containing carbon. As the particle containing carbon, for example, at least one selected from the group consisting of a carbon particle and a hybrid particle can be used, and a carbon particle is favorably used.

As the carbon particle, for example, at least one selected from the group consisting of carbon black, acetylene black, ketjen black, carbon nanotube, and graphene can be used, and carbon black of these carbon particles is favorably used. As the carbon black, for example, SEAST TA manufactured by Tokai Carbon Co., Ltd. or Asahi #15, #15HS, or the like manufactured by ASahi CARBON CO., LTD. can be used.

The hybrid particle contains carbon and a material other than carbon. The material other than carbon is, for example, an organic material or an inorganic material. The hybrid particle may be a hybrid particle in which carbon is attached to the surface of an inorganic particle. Specifically, for example, the hybrid particle may be hybrid carbon in which carbon is attached to the surface of a silica particle. (Second Particle)

As shown in Part B of FIG. 3, some particles of the second particle **432** included in the magnetic layer **43** protrude from the magnetic surface to form the protrusion **430**. When the head unit **56** slides over the magnetic tape MT, the protrusion **430** formed by the second particle **432** is capable of coming into contact with the head unit **56**.

The second particle **432** may be an abrasive. The lower limit value of the Mohs hardness of the second particle **432** is 7.0 or more, favorably 7.5 or more, more favorably 8.0 or more, and still more favorably 8.5 or more from the viewpoint of suppressing deformation due to contact with the head unit **56**. The upper limit value of the Mohs hardness of the second particle **432** is favorably 9.5 or less from the viewpoint of suppressing wear of the head unit **56**.

The second particle **432** is favorably an inorganic particles. Examples of the inorganic particle include α -alumina having an transformation rate of 90% or more, β -alumina, γ -alumina, silicon carbide, chromium oxide, cerium oxide, α -iron oxide, corundum, silicon nitride, titanium carbide, titanium oxide, silicon dioxide, tin oxide, magnesium oxide, tungsten oxide, zirconium oxide, boron nitride, zinc oxide, calcium carbonate, calcium sulfate, barium sulfate, molybdenum disulfide, acicular α -iron oxide obtained by dehydrating and annealing a raw material of magnetic iron oxide, those obtained by performing surface treatment thereon with aluminum and/or silica as necessary, and a diamond powder. As the inorganic particle, alumina particles such as α -alumina, β -alumina, and γ -alumina, and silicon carbide are favorably used. The second particle **432** may have any shape such as a needle shape, a spherical shape, and a dice shape, but those with some corners in their shapes are favorable because they have high abrasiveness. (Binder)

The binder includes, for example, a thermoplastic resin. The binder may further include a thermosetting resin, a reactive resin, or the like.

The thermoplastic resin includes a first thermoplastic resin containing a chlorine atom (first binder) and a second thermoplastic resin containing a nitrogen atom (second binder). More specifically, the thermoplastic resin includes a vinyl chloride resin and a urethane resin. In the present specification, the vinyl chloride resin means a polymer including a structural unit derived from vinyl chloride. More specifically, for example, the vinyl chloride resin means a homopolymer of vinyl chloride, a polymer of vinyl chloride and a comonomer that is copolymerizable therewith, and a mixture of these polymers.

The vinyl chloride resin includes, for example, at least one selected from the group consisting of vinyl chloride, a vinyl chloride-vinyl acetate copolymer, a vinyl chloride-vinylidene chloride copolymer, a vinyl chloride-acrylonitrile copolymer, an acrylic acid ester-vinyl chloride-vinylidene chloride copolymer, and a methacrylic acid ester-vinyl chloride copolymer.

The urethane resin means a resin including a urethane bond in at least part of the molecular chain constituting the resin, and may be a urethane resin or a copolymer including a urethane bond in part of the molecule chain. The urethane resin may be, for example, one obtained by reacting a

polyisocyanate with a polyol. Alternatively, the urethane resin may be, for example, one obtained by reacting a polyester with a polyol. In the present specification, the urethane resin also includes one obtained by a reaction with a curing agent.

The polyisocyanate includes, for example, at least one selected from the group consisting of diphenylmethane diisocyanate (MDI), tolylene diisocyanate (TDI), xylylene diisocyanate (XDI), 1,5-pentamethylene diisocyanate (PDI), hexamethylene diisocyanate (HDI), and isophorone diisocyanate (IPDI). In the present specification, the polyisocyanate means a compound having two or more isocyanate groups in the molecule. The polyisocyanate may be a polyisocyanate contained in a curing agent.

An arbitrary appropriate polyol can be adopted as the polyol as long as it is a polyol having two or more OH groups. The polyol includes, for example, at least one selected from the group consisting of a polyol having two OH groups (diol), a polyol having three OH groups (triol), a polyol having four OH groups (tetraol), a polyol having five OH groups (pentaol), and a polyol having six OH groups (hexaol). Specifically, the polyol includes, for example, at least one selected from the group consisting of a polyester polyol, a polyether polyol, a polycarbonate polyol, a polyesteramide polyol, and an acrylate polyol.

The polyester includes at least one selected from the group consisting of a phthalic acid polyester and an aliphatic polyester.

The thermoplastic resin may further include a thermoplastic resin other than the vinyl chloride resin and the urethane resin. Such a thermoplastic resin include, for example, at least one selected from the group consisting of vinyl acetate, an acrylic acid ester-acrylonitrile copolymer, an acrylic acid ester-acrylonitrile copolymer, an acrylic acid ester-vinylidene chloride copolymer, a methacrylic acid ester-vinylidene chloride copolymer, a methacrylic acid ester-ethylene copolymer, polyvinyl fluoride, a vinylidene chloride-acrylonitrile copolymer, an acrylonitrile-butadiene copolymer, a polyamide resin, polyvinyl butyral, a cellulose derivative (cellulose acetate butyrate, cellulose diacetate, cellulose triacetate, cellulose propionate, nitrocellulose), a styrene butadiene copolymer, a polyester resin, an amino resin, and synthetic rubber.

The thermosetting resin includes, for example, at least one selected from the group consisting of a phenolic resin, an epoxy resin, a polyurethane curable resin, a urea resin, a melamine resin, an alkyd resin, a silicone resin, a polyamine resin, and a urea formaldehyde resin.

In order to improve the dispersibility of the magnetic particles, a polar functional group such as $-\text{SO}_3\text{M}$, $-\text{OSO}_3\text{M}$, $-\text{COOM}$, $\text{P}=\text{O}(\text{OM})_2$ (where M represents a hydrogen atom or an alkali metal such as lithium, potassium, and sodium), a side chain amine having a terminal group represented by $-\text{NR}_1\text{R}_2$ or $-\text{NR}_1\text{R}_2\text{R}_3^+\text{X}^-$, a main chain amine represented by $>\text{NR}_1\text{R}_2^+\text{X}^-$ (where R₁, R₂, and R₃ each represent a hydrogen atom or a hydrocarbon group, and X⁻ represents a halogen element ion such as fluorine, chlorine, bromine, and iodine, an inorganic ion, or an organic ion.), $-\text{OH}$, $-\text{SH}$, $-\text{CN}$, or an epoxy group may be introduced into all the above binders. The amount of polar functional groups introduced into the binder is favorably 10^{-1} or more and 10^{-8} or less mol/g, more favorably 10^{-2} or more and 10^{-6} or less mol/g. (Lubricant)

The lubricant contains, for example, at least one of a fatty acid or a fatty acid ester, favorably both a fatty acid and a fatty acid ester. The magnetic layer **43** containing a lubri-

cant, particularly the magnetic layer **43** containing both a fatty acid and a fatty acid ester, contributes to improving the travelling stability the magnetic tape MT. More particularly, when the magnetic layer **43** contains a lubricant and has a pore, favorable travelling stability can be achieved. The reason why the travelling stability is improved is presumably because the dynamic friction coefficient of the surface of the magnetic tape MT on the side of the magnetic layer **43** is adjusted to a value suitable for travelling of the magnetic tape MT by the above lubricant.

The fatty acid may favorably be a compound represented by the following general formula (1) or (2). For example, one of the compound represented by the following general formula (1) and the compound represented by the general formula (2) may be contained as a fatty acid, or both of them may be contained.

Further, the fatty acid ester may favorably be a compound represented by the following general formula (3) or (4). For example, one of the compound represented by the following general formula (3) and the compound represented by the general formula (4) may be contained as the fatty acid ester, or both of them may be contained.

When the lubricant contains one or both of the compound represented by the general formula (1) and the compound represented by the general formula (2) and the compound represented by the general formula (3) and the compound represented by the general formula (4), it is possible to suppress an increase in the dynamic friction coefficient due to repeated recording or reproduction of the magnetic tape MT.



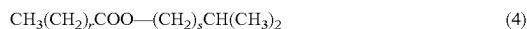
(However, in the general formula (1), k represents an integer selected from the range of 14 or more and 22 or less, more favorably the range of 14 or more and 18 or less.)



(However, in the general formula (2), the sum of n and m is an integer selected from the range of 12 or more and 20 or less, more favorably the range of 14 or more and 18 or less.)



(However, in the general formula (3), p represents an integer selected from the range of 14 or more and 22 or less, more favorably the range of 14 or more and 18 or less, and q represents an integer selected from the range of 2 or more and 5 or less, more favorably the range of 2 or more and 4 or less.)



(However, in the general formula (4), r represents an integer selected from the range of 14 or more and 22 or less, and s represents an integer selected from the range of 1 or more and 3 or less.)
(Antistatic Agent)

The antistatic agent includes a carbon particle. The antistatic agent may further include at least one selected from the group consisting of a natural surfactant, a nonionic surfactant, and a cationic surfactant. The carbon particle includes, for example, at least one selected from the group consisting of carbon black, acetylene black, ketjen black, carbon nanotube, and graphene.
(Curing Agent)

The curing agent includes, for example, a polyisocyanate. The polyisocyanate may be, for example, one containing, as an isocyanate source, diphenylmethane diisocyanate (MDI),

tolylene diisocyanate (TDI), xylylene diisocyanate (XDI), 1,5-pentamethylene diisocyanate (PDI), hexamethylene diisocyanate (HDI), or isophorone diisocyanate (IPDI). The polyisocyanate may have a TMP adduct structure, an isocyanurate structure, a biuret structure, an allophanate structure, or the like.

Specifically, the polyisocyanate includes, for example, aromatic polyisocyanate such as an adduct of tolylene diisocyanate (TDI) and an active hydrogen compound, or aliphatic polyisocyanate such as an adduct of hexamethylene diisocyanate (HMDI) and an active hydrogen compound. The weight average molecular weight of these polyisocyanates is desirably within the range of 100 or more and 3000 or less.

(Rust Inhibitor)

Examples of the rust inhibitor include phenols, naphthols, quinones, a heterocyclic compound containing a nitrogen atom, a heterocyclic compound containing an oxygen atom, and a heterocyclic compound containing a sulfur atom.

(Non-Magnetic Reinforcing Particle)

Examples of the non-magnetic reinforcing particles include aluminum oxide (α , β , or γ alumina), chromium oxide, silicon oxide, diamond, garnet, emery, boron nitride, titanium carbide, silicon carbide, titanium carbide, and titanium oxide (rutile or anatase type titanium oxide).

(Underlayer)

The underlayer **42** is for reducing the unevenness of the surface of the base **41** and adjusting the unevenness of the magnetic surface. The underlayer **42** is a non-magnetic layer including non-magnetic particles, a binder, and a lubricant. The underlayer **42** supplies the lubricant to the magnetic surface. The underlayer **42** may further include at least one additive selected from the group consisting of an antistatic agent, a curing agent, and a rust inhibitor, as necessary.

An average thickness t_2 of the underlayer **42** is favorably 300 nm or more and 1200 nm or less, more favorably 300 nm or more and 900 nm or less, 300 nm or more and 600 nm or less. Note that the average thickness t_2 of the underlayer **42** is obtained in the same manner as the average thickness t_1 of the magnetic layer **43**. However, the magnification of the TEM image is adjusted as appropriate in accordance with the thickness of the underlayer **42**. When the average thickness t_2 of the underlayer **42** is 1200 nm or less, the stretchability of the magnetic tape MT due to an external force further increases, and thus, the width of the magnetic tape MT can be adjusted more easily by tension adjustment.

The underlayer **42** favorably includes a plurality of hole portions. By storing the lubricant in the plurality of hole portions, it is possible to further suppress a decrease in the amount of lubricant supplied between the magnetic surface and the head unit **56** even after repeated recording or reproduction (i.e., after the head unit **56** is caused to come into contact with the surface of the magnetic tape MT and repeatedly travel). Therefore, it is possible to further suppress an increase in the dynamic friction coefficient. That is, further excellent travelling stability can be achieved.
(Non-Magnetic Particles)

The non-magnetic particles include, for example, at least one of inorganic particles or organic particles. Further, the non-magnetic particles may be carbon particles such as carbon black. Note that one type of non-magnetic particle may be used alone, or two or more types of non-magnetic particles may be used in combination. The inorganic particles contain, for example, a metal, a metal oxide, a metal carbonate, a metal sulfate, a metal nitride, a metal carbide, a metal sulfide, or the like. Examples of the shape of the non-magnetic particles include various shapes such as a

needle shape, a spherical shape, a cubic shape, and a plate shape, but the shape is not limited thereto.

(Binder and Lubricant)

The binder and the lubricant are similar to those in the above magnetic layer 43.

(Additive)

The antistatic agent, the curing agent, and the rust inhibitor are similar to those in the above magnetic layer 43.

(Back Layer)

The back layer 44 includes a binder and non-magnetic particles. The back layer 44 may further include at least one additive selected from the group consisting of a lubricant, a curing agent, and an antistatic agent, as necessary. The binder and the non-magnetic particles are similar to those in the above underlayer 42. The curing agent and the antistatic agent are similar to those in the above magnetic layer 43.

The average particle size of the non-magnetic particles is favorably 10 nm or more and 150 nm or less, more favorably 15 nm or more and 110 nm or less. The average particle size of the non-magnetic particles is obtained in the same manner as the above average particle size of the magnetic particles. The non-magnetic particles may include non-magnetic particles having two or more granularity distributions.

The upper limit value of the average thickness of the back layer 44 is favorably 0.60 μm or less. When the upper limit value of the average thickness of the back layer 44 is 0.60 μm or less, the underlayer 42 and the base 41 can be kept thick even in the case where the average thickness of the magnetic tape MT is 5.60 μm or less, and thus, the travelling stability of the magnetic tape MT in the recording/reproduction apparatus can be maintained. The lower limit value of the average thickness of the back layer 44 is not particularly limited, but is, for example, 0.20 μm or more.

An average thickness t_b of the back layer 44 is obtained as follows. First, an average thickness t_T of the magnetic tape MT is measured. The measurement method of the average thickness t_T is as described in the following "Average thickness of magnetic tape". Subsequently, the magnetic tape MT housed in the cartridge 10 is unwound, and the magnetic tape MT is cut into a length of 250 mm at a position 30 m to 40 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT to prepare a sample. Next, the back layer 44 of the sample is removed with a solvent such as MEK (methyl ethyl ketone) and dilute hydrochloric acid. Next, the thickness of the sample is measured at five positions using a Laser Hologage (LGH-110C) manufactured by Mitutoyo Corporation, and the measured values are simply averaged (arithmetically averaged) to calculate an average value t_b [μm]. After that, the average thickness t_b [μm] of the back layer 44 is obtained in accordance with the following formula. Note that the above five measurement positions are randomly selected from the sample such that they are different positions in the longitudinal direction of the magnetic tape MT.

$$t_b[\mu\text{m}] = t_T[\mu\text{m}] - t_B[\mu\text{m}]$$

(Average Thickness of Magnetic Tape)

The upper limit value of the average thickness (average total thickness) t_T of the magnetic tape MT is favorably 5.30 μm or less, more favorably 5.10 μm or less, still more favorably 4.90 μm or less, and particularly favorably 4.70 μm or less. When the average thickness t_T of the magnetic tape MT is 5.30 μm or less, it is possible to increase the

recording capacity of one data cartridge to be more than that of a general magnetic tape. The lower limit value of the average thickness t_T of the magnetic tape MT is not particularly limited, but is, for example, 3.50 μm or more.

The average thickness t_T of the magnetic tape MT is obtained as follows. First, the magnetic tape MT housed in the cartridge 10 is unwound, and the magnetic tape MT is cut into a length of 250 mm at a position 30 m to 40 m in the longitudinal direction from a connection part 21 between the magnetic tape MT and the leader tape LT to prepare a sample. Next, the thickness of the sample is measured at five positions using a Laser Hologage (LGH-110C) manufactured by Mitutoyo Corporation, and the measured values are simply averaged (arithmetically averaged) to calculate an average thickness t_T [μm]. Note that the above five measurement positions are randomly selected from the sample such that they are different positions in the longitudinal direction of the magnetic tape MT.

(Coercive Force Hc2)

The upper limit value of the coercive force Hc2 of the magnetic layer 43 in the longitudinal direction of the magnetic tape MT is favorably 2000 Oe or less, more favorably 1900 Oe or less, and still more favorably 1800 Oe or less. When the coercive force Hc2 of the magnetic layer 43 in the longitudinal direction of the magnetic tape MT is 2000 Oe or less, sufficient electromagnetic conversion characteristics can be provided even with high recording density.

The lower limit value of the coercive force Hc2 of the magnetic layer 43 measured in the longitudinal direction of the magnetic tape MT is favorably 1000 Oe or more. When the coercive force Hc2 of the magnetic layer 43 measured in the longitudinal direction of the magnetic tape MT is 1000 Oe or more, it is possible to suppress demagnetization due to leakage flux from the recording head.

The above coercive force Hc2 is obtained as follows. First, the magnetic tape MT housed in the cartridge 10 is unwound, the magnetic tape MT is cut out at a position 30 m to 40 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT, and three magnetic tapes MT are stacked with double sided tape such that orientations of the magnetic tapes MT in the longitudinal direction are the same and then punched out with a $\phi 6.39$ mm punch to prepare a measurement sample. At this time, marking is performed with arbitrary non-magnetic ink such that the longitudinal direction (travelling direction) of the magnetic tape MT can be recognized. Then, the M-H loop of the measurement sample (entire magnetic tape MT) corresponding to the longitudinal direction (travelling direction) of the magnetic tape MT is measured using a vibrating sample magnetometer (VSM). Next, the coating films (the underlayer 42, the magnetic layer 43, the back layer 44, and the like) of the magnetic tape MT cut as described above are wiped off with acetone, ethanol, or the like, leaving only the base 41. Then, three obtained bases 41 are stacked with double sided tape and then punched out with a $\phi 6.39$ mm punch to prepare a sample for background correction (hereinafter referred to simply as a "correction sample"). After that, the M-H loop of the correction sample (base 41) corresponding to the longitudinal direction of the base 41 (the longitudinal direction of the magnetic tape MT) is measured using a VSM.

A highly sensitive vibrating sample magnetometer "VSM-P7-15" manufactured by TOEI INDUSTRY CO., LTD. is used to measure the M-H loop of the measurement sample (the entire magnetic tape MT) and the M-H loop of the correction sample (base 41). The measurement conditions are the measurement mode: full-loop, the maximum

magnetic field: 15 kOe, the magnetic field step: 40 bits, the time constant of locking amp: 0.3 sec, the waiting time: 1 sec, and the MH average number: 20.

After obtaining the M-H loop of the measurement sample (the entire magnetic tape MT) and the M-H loop of the correction sample (base **41**), the M-H loop of the correction sample (base **41**) is subtracted from the M-H loop of the measurement sample (the entire magnetic tape MT) to perform background correction, thereby obtaining the M-H loop after background correction. The measurement/analysis program attached to the "VSM-P7-15" is used for this calculation of background correction. The coercive force Hc2 is obtained on the basis of the obtained M-H loop after background correction. Note that for this calculation, the measurement/analysis program attached to the "VSM-P7-15" is used. Note that the above measurement of M-H loops is performed at 25° C.±2° C. and 50% RH±5% RH. Further, "demagnetizing field correction" when measuring the M-H loop in the longitudinal direction of the magnetic tape MT is not performed.

(Squareness Ratio)

A squareness ratio S1 of the magnetic layer **43** in the perpendicular direction of the magnetic tape MT is favorably 62% or more, more favorably 65% or more, still more favorably 68% or more, particularly favorably 72% or more, and most favorably 75% or more. When the squareness ratio S1 is 62% or more, the perpendicular orientation of the magnetic particles is sufficiently high, and thus, further excellent electromagnetic conversion characteristics can be achieved.

The squareness ratio S1 in the perpendicular direction of the magnetic tape MT is obtained as follows. First, the magnetic tape MT housed in the cartridge **10** is unwound, the magnetic tape MT is cut out at a position 30 m to 40 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT, and three magnetic tapes MT are stacked with double sided tape such that orientations of the magnetic tapes MT in the longitudinal direction are the same and then punched out with a φ6.39 mm punch to prepare a measurement sample. At this time, marking is performed with arbitrary non-magnetic ink such that the longitudinal direction (travelling direction) of the magnetic tape MT can be recognized. Then, the M-H loop of the measurement sample (entire magnetic tape MT) corresponding to the perpendicular direction of the magnetic tape MT (the perpendicular direction of the magnetic tape MT) is measured using a vibrating sample magnetometer (VSM). Next, the coating films (the underlayer **42**, the magnetic layer **43**, the back layer **44**, and the like) of the magnetic tape MT cut as described above are wiped off with acetone, ethanol, or the like, leaving only the base **41**. Then, three obtained bases **41** are stacked with double sided tape and then punched out with a φ6.39 mm punch to prepare a sample for background correction (hereinafter referred to simply as a "correction sample"). After that, the M-H loop of the correction sample (base **41**) corresponding to the perpendicular direction of the base **41** (the perpendicular direction of the magnetic tape MT) is measured using a VSM.

A highly sensitive vibrating sample magnetometer "VSM-P7-15" manufactured by TOEI INDUSTRY CO., LTD. is used to measure the M-H loop of the measurement sample (the entire magnetic tape MT) and the M-H loop of the correction sample (base **41**). The measurement conditions are the measurement mode: full-loop, the maximum magnetic field: 15 kOe, the magnetic field step: 40 bits, the

time constant of locking amp: 0.3 sec, the waiting time: 1 sec, and the MH average number: 20.

After obtaining the M-H loop of the measurement sample (the entire magnetic tape MT) and the M-H loop of the correction sample (base **41**), the M-H loop of the correction sample (base **41**) is subtracted from the M-H loop of the measurement sample (the entire magnetic tape MT) to perform background correction, thereby obtaining the M-H loop after background correction. The measurement/analysis program attached to the "VSM-P7-15" is used for this calculation of background correction.

A saturation magnetization Ms (emu) and a residual magnetization Mr (emu) of the obtained M-H loop after background correction are substituted into the following formula to calculate the squareness ratio S1(%). Note that the above measurement of M-H loops is performed at 25° C.±2° C. and 50% RH±5% RH. Further, "demagnetizing field correction" when measuring the M-H loop in the perpendicular direction of the magnetic tape MT is not performed. Note that for this calculation, the measurement/analysis program attached to the "VSM-P7-15" is used.

$$\text{Squareness ratio } S1 (\%) = (Mr/Ms) \times 100$$

A squareness ratio S2 of the magnetic layer **43** in the longitudinal direction (travelling direction) of the magnetic tape MT is favorably 35% or less, more favorably 30% or less, still more favorably 25% or less, particularly favorably 20% or less, and most favorably 15% or less. When the squareness ratio S2 is 35% or less, the perpendicular orientation of the magnetic particles is sufficiently high, and thus, further excellent electromagnetic conversion characteristics can be achieved. Note that one of the squareness ratio S1 of the magnetic layer **43** in the perpendicular direction of the magnetic tape MT and the squareness ratio S2 of the magnetic layer **43** in the longitudinal direction (travelling direction) of the magnetic tape MT may be within the above favorable range, and the other may be outside the above favorable range. Alternatively, both the squareness ratio S1 of the magnetic layer **43** in the perpendicular direction of the magnetic tape MT and the squareness ratio S2 of the magnetic layer **43** in the longitudinal direction (travelling direction) of the magnetic tape MT may be within the above favorable range.

The squareness ratio S2 in the longitudinal direction of the magnetic tape MT is obtained in the same manner as the squareness ratio S1 except that the M-H loop is measured in the longitudinal direction (travelling direction) of the magnetic tape MT and the base **41**.

(Ratio Hc2/Hc1)

A ratio Hc2/Hc1 of a coercive force Hc1 of the magnetic layer **43** in the perpendicular direction of the magnetic tape MT and the coercive force Hc2 of the magnetic layer **43** in the longitudinal direction of the magnetic tape MT satisfies the relationship of favorably Hc2/Hc1≤0.8, more favorably Hc2/Hc1≤0.75, still more favorably Hc2/Hc1≤0.7, particularly favorably Hc2/Hc1≤0.65, and most favorably Hc2/Hc1≤0.6. When the coercive forces Hc1 and Hc2 satisfy the relationship of Hc2/Hc1≤0.8, it is possible to increase the degree of perpendicular orientation of the magnetic particles. Therefore, since the magnetization transition width can be reduced and a high-output signal can be obtained during signal reproduction, further excellent electromagnetic conversion characteristics can be achieved. Note that since the magnetization reacts sensitively to the magnetic

field in the perpendicular direction from the recording head when Hc2 is small, as described above, a favorable recording pattern can be formed.

In the case where the ratio Hc2/Hc1 satisfies the relationship of $Hc2/Hc1 \leq 0.8$, it is particularly effective that the average thickness t_1 of the magnetic layer 43 is 90 nm or less. When the average thickness t_1 of the magnetic layer 43 exceeds 90 nm, in the case of using a ring-type head as a recording head, there is a possibility that a lower region of the magnetic layer 43 (region on the side of the underlayer 42) is magnetized in the longitudinal direction of the magnetic tape MT and the magnetic layer 43 cannot be uniformly magnetized in the thickness direction. Therefore, even in the case where the ratio Hc2/Hc1 satisfies the relationship of $Hc2/Hc1 \leq 0.8$ (i.e., the degree of perpendicular orientation of the magnetic particles is increased), there is a possibility that further excellent electromagnetic conversion characteristics cannot be achieved.

The lower limit value of Hc2/Hc1 is not particularly limited, but satisfies the following relationship: $0.5 \leq Hc2/Hc1$, for example. Note that Hc2/Hc1 represents the degree of perpendicular orientation of the magnetic particles, and the smaller Hc2/Hc1, the higher the degree of perpendicular orientation of the magnetic particles.

The method of calculating the coercive force Hc2 of the magnetic layer 43 in the longitudinal direction of the magnetic tape MT is as described above. The coercive force Hc1 of the magnetic layer 43 in the perpendicular direction of the magnetic tape MT is obtained in the same manner as the coercive force Hc2 of the magnetic layer 43 in the longitudinal direction of the magnetic tape MT except that the M-H loop is measured in the perpendicular direction (thickness direction) of the magnetic tape MT and the base 41. (Activation Volume V_{act})

The activation volume V_{act} is favorably 8000 nm³ or less, more favorably 6000 nm³ or less, still more favorably 5000 nm³ or less, particularly favorably 4000 nm³ or less, and most favorably 3000 nm³ or less. When the activation volume V_{act} is 8000 nm³ or less, the dispersion state of the magnetic particles is favorable, which allows the bit-inversion region to be steep and makes it possible to prevent the magnetization signal recorded on the adjacent track from deteriorating due to the leakage magnetic field from the recording head. Therefore, there is a possibility that further excellent electromagnetic conversion characteristics cannot be achieved.

The above activation volume V_{act} is obtained by the following formula derived by Street&Woolley.

$$V_{act}(\text{nm}^3) = k_B \times T \times X_{irr}/(\mu_0 \times Ms \times S)$$

(where k_B : Boltzmann constant (1.38×10^{-23} J/K), T : temperature (K), X_{irr} : irreversible magnetic susceptibility, μ_0 : vacuum permeability, S : magnetic viscosity coefficient, Ms : saturation magnetization (emu/cm³))

The irreversible magnetic susceptibility X_{irr} , the saturation magnetization Ms , and the magnetic viscosity coefficient S substituted into the above formula are obtained as follows using a VSM. Note that the measurement direction using a VSM is the perpendicular direction (thickness direction) of the magnetic tape MT. Further, measurement using a VSM is performed on the measurement sample cut from the long magnetic tape MT at 25° C $\pm$ 2° C. and 50% RH $\pm$ 5% RH. Further, "demagnetizing field correction" when mea-

suring the M-H loop in the perpendicular direction (thickness direction) of the magnetic tape MT is not performed. (Irreversible Magnetic Susceptibility X_{irr})

The irreversible magnetic susceptibility X_{irr} is defined as the slope of the residual magnetization curve (DCD curve) near a residual coercive force H_r . First, a magnetic field of -1193 kA/m (15 kOe) is applied to the entire magnetic tape MT, and the magnetic field is returned to zero to create a residual magnetization state. After that, a magnetic field of approximately 15.9 kA/m (200 Oe) is applied in the opposite direction, and the magnetic field is returned to zero again to measure the residual magnetization amount. After that, measurement in which a magnetic field 15.9 kA/m larger than the previously applied magnetic field is applied and the magnetic field is returned to zero is repeatedly performed similarly, and the residual magnetization amount is plotted against the applied magnetic field to measure the DCD curve. The point where the magnetization amount is zero in the obtained DCD curve is used as the residual coercive force H_r , and the DCD curve is differentiated to obtain the slope of the DCD curve in each magnetic field. Of the slopes of the DCD curve, the slope near the residual coercive force H_r is X_{irr} .

(Saturation Magnetization Ms)

First, the M-H loop after background correction is obtained in the same manner as the above measurement method of the squareness ratio $S1$. Next, Ms (emu/cm³) is calculated from the value of the saturation magnetization Ms (emu) of the obtained M-H loop and the volume (cm³) of the magnetic layer 43 in the measurement sample. Note that the volume of the magnetic layer 43 is obtained by multiplying the area of the measurement sample by the average thickness t_1 of the magnetic layer 43. The method of calculating the average thickness t_1 of the magnetic layer 43 necessary for calculating the volume of the magnetic layer 43 is as described above.

(Magnetic Viscosity Coefficient S)

First, a magnetic field of -1193 kA/m (15 kOe) is applied to the entire magnetic tape MT (measurement sample), and the magnetic field is returned to zero to create a residual magnetization state. After that, a magnetic field equivalent to the value of the residual coercive force H_r obtained from the DCD curve is applied in the opposite direction. The magnetization amount is continuously measured at predetermined intervals for 1000 seconds while applying the magnetic field. The relationship between the time t and the magnetization amount $M(t)$ obtained in this way is referred to in the following formula to calculate the magnetic viscosity coefficient S .

$$M(t) = MO + S \times \ln(t)$$

(where $M(t)$: magnetization amount at the time t , MO : initial magnetization amount, S : magnetic viscosity coefficient, $\ln(t)$: natural logarithm of time) (Surface Roughness R_b of Back Surface)

A surface roughness of the back surface (surface roughness of the back layer 44) R_b favorably satisfies the relationship of $R_b \leq 6.0$ [nm]. When the surface roughness R_b of the back surface is within the above range, further excellent electromagnetic conversion characteristics can be achieved.

The surface roughness R_b of the back surface is obtained as follows. First, the magnetic tape MT housed in the cartridge 10 is unwound, and the magnetic tape MT is cut into a length of 100 mm at a position 30 m to 40 m in the

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longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT to prepare a sample. Next, the sample is placed on a slide glass such that the surface thereof to be measured (surface on the side of the magnetic layer) faces upward, and the end portion of the sample is fixed with mending tape. The surface shape is measured using VertScan (objective lens 20×) as a measuring apparatus, and the surface roughness R_b of the back surface is obtained from the following formula on the basis of the ISO 25178 standard.

The measurement conditions are as follows.

Apparatus: non-contact profilometer using optical interference

(non-contact surface/layer cross-sectional shape measurement system VertScan R5500GL-M100-AC manufactured by Ryoka Systems Inc.)

Objective lens: 20×

Measurement region: 640×480 pixels (field of view: approximately 237 μm×178 μm field of view)

Measurement mode: phase

Wavelength filter: 520 nm

CCD: 1/3 inch

Noise removal filter: smoothing 3×3

Surface correction: correction using a quadratic polynomial approximation surface

Measurement software: VS-Measure Version 5.5.2

Analysis software: VS-viewer Version 5.5.5

$$S_a = \frac{1}{A} \iint_A |Z(x, y)| dx dy$$

After measuring the surface roughness at five positions in the longitudinal direction of the magnetic tape MT as described above, the average value of arithmetic average roughnesses S_a (nm) automatically calculated from the surface profiles obtained at the respective positions is used as the surface roughness R_b of the back surface (nm). (Young's Modulus in Longitudinal Direction of Magnetic Tape)

The upper limit value of the Young's modulus in the longitudinal direction of the magnetic tape MT is favorably 9.0 GPa or less, more favorably 8.0 GPa or less, still more favorably 7.5 GPa or less, and particularly favorably 7.1 GPa or less. When the Young's modulus in the longitudinal direction of the magnetic tape MT is 9.0 GPa or less, the stretchability of the magnetic tape MT due to an external force further increases, and thus, the width of the magnetic tape MT can be adjusted more easily by tension adjustment. Therefore, it is possible to more appropriately suppress off-track and more accurately reproduce the data recorded on the magnetic tape MT. The lower limit value of the Young's modulus in the longitudinal direction of the magnetic tape MT is favorably 3.0 GPa or more, more favorably 4.0 GPa or more. When the lower limit value of the Young's modulus in the longitudinal direction of the magnetic tape MT is 3.0 GPa or more, it is possible to suppress a decrease in travelling stability.

The Young's modulus in the longitudinal direction of the magnetic tape MT is a value indicating the difficulty of expansion and contraction in the longitudinal direction of the magnetic tape MT due to an external force. The larger this value, the more difficult it is for the magnetic tape MT to expand and contract in the longitudinal direction, and the

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smaller this value, the more easily the magnetic tape MT expands and contracts in the longitudinal direction due to an external force.

Note that although the Young's modulus in the longitudinal direction of the magnetic tape MT is a value relating to the longitudinal direction of the magnetic tape MT, it also correlates with the difficulty of expansion and contraction in the width direction of the magnetic tape MT. That is, the larger this value, the more difficult it is for the magnetic tape MT to expand and contract in the width direction due to an external force, and the smaller this value, the more easily the magnetic tape MT expands and contracts in the width direction due to an external force. Therefore, from the viewpoint of tension adjustment, it is advantageous for the Young's modulus in the longitudinal direction of the magnetic tape MT to be small as described above, i.e., 9.0 GPa or less.

A tensile tester (AG-100D manufactured by Shimadzu Corporation) is used to measure the Young's modulus. In the case where it is desired to measure the Young's modulus in the tape longitudinal direction, the magnetic tape MT housed in the cartridge **10** is unwound, and the magnetic tape MT is cut into a length of 180 mm at a position 30 m to 40 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT to prepare a measurement sample. A jig capable of fixing the width of the tape (1/2 inch) is attached to the above tensile tester to fix the top and bottom of the tape width. The distance (length of the tape between chucks) is set to 100 mm. After the tape sample is chucked, stress is gradually applied in the direction in which the sample is pulled. The pulling speed is set to 0.1 mm/min. From the change in the stress and the amount of elongation at this time, the Young's modulus is calculated using the following formula.

$$E(N/m^2) = ((\Delta N/S)/(\Delta x/L)) \times 10^6$$

ΔN : change in stress (N)

S : cross-sectional area of test piece (mm²)

Δx : amount of elongation (mm)

L : distance between gripping jigs (mm)

The cross-sectional area S of the above measurement sample **10S** is the cross-sectional area before the tensile operation, and is obtained as the product of the width of the measurement sample **10S** (1/2 inch) and the thickness of the measurement sample **10S**. As the range of tensile stress when performing measurement, a range of tensile stress of a linear region is set in accordance with the thickness of the magnetic tape MT or the like. Here, the stress range is set to 0.2 N to 0.7 N, and the change in stress (ΔN) and the amount of elongation (Δx) at this time are used for calculation. Note that the above measurement of the Young's modulus is performed at 25° C.±2° C. and 50% RH±5% RH.

(Young's Modulus in Longitudinal Direction of Base)

The Young's modulus in the longitudinal direction of the base **41** is favorably 7.8 GPa or less, more favorably 7.0 GPa or less, still more favorably 6.6 GPa or less, and particularly favorably 6.4 GPa or less. When the Young's modulus in the longitudinal direction of the base **41** is 7.8 GPa or less, the stretchability of the magnetic tape MT due to an external force further increases, and thus, the width of the magnetic tape MT can be adjusted more easily by tension adjustment. Therefore, it is possible to more appropriately suppress off-track and more accurately reproduce the data recorded on the magnetic tape MT. The lower limit value of the

Young's modulus in the longitudinal direction of the base **41** is favorably 2.5 GPa or more, more favorably 3.0 GPa or more. When the lower limit value of the Young's modulus in the longitudinal direction of the base **41** is 2.5 GPa or more, it is possible to suppress a decrease in travelling stability.

The above Young's modulus in the longitudinal direction of the base **41** is obtained as follows. First, the magnetic tape MT housed in the cartridge **10** is unwound, and the magnetic tape MT is cut into a length of 180 mm at a position 30 m to 40 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT. Subsequently, the underlayer **42**, the magnetic layer **43**, and the back layer **44** are removed from the cut magnetic tape MT to obtain the base **41**. The Young's modulus in the longitudinal direction of the base **41** is obtained using this base **41** in the same procedure as that for the above Young's modulus in the longitudinal direction of the magnetic tape MT.

The thickness of the base **41** occupies more than half of the thickness of the entire magnetic tape MT. Therefore, the Young's modulus in the longitudinal direction of the base **41** correlates with the difficulty of expansion and contraction of the magnetic tape MT due to an external force. The larger this value, the more difficult it is for the magnetic tape MT to expand and contract in the width direction due to an external force, and the smaller this value, the more easily the magnetic tape MT expands and contracts in the width direction due to an external force.

Note that although the Young's modulus in the longitudinal direction of the base **41** is a value relating to the longitudinal direction of the magnetic tape MT, it also correlates with the difficulty of expansion and contraction in the width direction of the magnetic tape MT. That is, the larger this value, the more difficult it is for the magnetic tape MT to expand and contract in the width direction due to an external force, and the smaller this value, the more easily the magnetic tape MT expands and contracts in the width direction due to an external force. Therefore, from the viewpoint of tension adjustment, it is advantageous for the Young's modulus in the longitudinal direction of the base **41** to be small as described above, i.e., 7.8 GPa or less. (Atomic Concentration Ratio (A2/A1) of Magnetic Layer **43**)

An atomic concentration ratio (A2/A1) of a maximum chlorine atomic concentration A1 obtained by measuring the chlorine atomic concentration of the magnetic layer **43** in the thickness direction of the magnetic layer **43** and a maximum nitrogen atomic concentration A2 obtained by measuring the nitrogen atomic concentration of the magnetic layer **43** in the thickness direction of the magnetic layer **43** is 0.50 or more and 0.75 or less, favorably 0.55 or more and 0.71 or less. When the atomic concentration ratio (A2/A1) is less than 0.50, since the content of the vinyl chloride resin relative to the content of the urethane resin is too high, the surface of the magnetic layer **43** becomes too hard and the surface of the magnetic layer **43** becomes susceptible to brittle fracture when the head unit **56** slides over the magnetic tape MT. That is, the protrusion **430** on the surface of the magnetic layer **43** is scraped off, and powder fall occurs easily during recording or reproduction. Therefore, there is a possibility that the friction between the magnetic tape MT and the head unit **56** increases during recording or reproduction. Meanwhile, when the atomic concentration ratio (A2/A1) exceeds 0.75, since the content of the vinyl chloride resin relative to the content of the urethane resin is too small, the surface of the magnetic layer **43** becomes too soft and the

protrusion **430** on the magnetic surface is easily plastically deformed by travelling of the magnetic tape MT. Therefore, there is a possibility that the friction between the magnetic tape MT and the head unit **56** increases during recording or reproduction. Note that the chlorine atomic concentration is derived from the content of the vinyl chloride resin contained in the magnetic layer **43**, and the nitrogen atomic concentration is derived from the content of the urethane resin contained in the magnetic layer **43**.

The atomic concentration ratio (A2/A1) is obtained as follows.

(Method of Preparing Sample of STEM Observation)

First, the magnetic tape MT housed in the cartridge **10** is unwound, and a sample is cut out from a position 20 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT. The sample is cut out into an appropriate size (e.g., a rectangle of approximately 1 mm×approximately 1 mm) from substantially the central portion in the width direction of the magnetic tape MT. Next, the magnetic surface of the cut sample is subjected to carbon deposition treatment to form a carbon layer on the magnetic surface. After that, the sample is introduced into a focused-ion beam (FIB) processing apparatus (FIB-SEM apparatus) equipped with a scanning electron microscope (SEM).

Next, the sample is micro-sampled to a size suitable for STEM observation (e.g., a rectangle with one side of 10 μm to 50 μm) using the focused-ion beam processing apparatus, thereby obtaining a microsample. Next, the microsample is fixed to the sample stage of the focused-ion beam processing apparatus and thinned. The thinning is performed such that the thickness of the microsample in the direction horizontal to the magnetic surface is a thickness that an electron beam used for STEM observation can be transmitted. In this way, a microsample for STEM observation is prepared. Note that the preparation apparatus of the microsample and the processing conditions are as follows.

[Apparatus]

FIB-SEM apparatus: Versa3D DualBeam manufactured by FEI

[Processing Conditions]

Pre-processing on sample: carbon deposition

Ion species: gallium ion

Acceleration voltage (voltage applied when roughly drilling): 30 kV

Final finishing voltage: 5 kV

Material of sample stage: Mo

(EDX Analysis and STEM Observation)

STEM observation and EDX analysis are performed on the microsample prepared as described above. The apparatuses used for STEM observation and EDX analysis, the conditions for STEM observation, and the conditions for EDX analysis are as follows.

[Apparatus]

STEM apparatus: TalosF200X (Schottky-FEG) manufactured by FEI

EDX system: Super-X manufactured by FEI

EDX detector: four windowless SDD detectors (30 mm², built-in objective lens) manufactured by Bruker

[STEM Observation Conditions]

Acceleration voltage: 200 kV

Acquired image: BF STEM image (Bright Field: BF)

HAADF STEM image (High Angle Annular Dark Field: HAADF)

Camera length: 98 mm

Apparatus display magnification: 57000 times

[EDX Analysis Conditions (Conditions for Mapping and Line Extraction in EDX Analysis)]

Acceleration voltage: 200 kV

Apparatus display magnification: 57000 times

Surface analysis resolution: 800 pixel×700 pixel (pixel size: approximately 2.1 nm)

Moving average filter: 3 pixel

Data type: net count

Line extraction integration width×length: 700 pixel×650 pixel

Cl is the number of K α ray counts.

N is the number of K α ray counts.

In the above EDX analysis conditions, the “Line extraction integration width” means a length L_a of a side substantially horizontal to the magnetic surface of the magnetic tape MT, of a rectangular region R where characteristic X-rays are extracted, as shown in FIG. 10. The “length in the thickness direction” means a length L_b of a side substantially horizontal to the thickness direction of the magnetic tape MT, of the rectangular region R where characteristic X-rays are extracted. Note that the direction of extraction of characteristic X-rays is a direction indicated by an arrow D in FIG. 10.

Under the above observation conditions, a HAADF STEM image of the above sample for STEM observation in a direction horizontal to the magnetic surface is acquired. An example of the acquired HAADF STEM image is shown in FIG. 10. As shown in the figure, the magnetic layer 43 and the underlayer 42 can be observed from the image. In this observation, the STEM cross-sectional photograph is checked to confirm that there are no abnormal portions in a state clearly different from the normal state of the magnetic layer 43 in the cross section of the magnetic layer 43. The abnormal portions are, for example, coarse inorganic particles, gaps, or undispersed binders, and EDX analysis is performed using the cross section where the above abnormal portions are not present.

By performing EDX analysis under the above analysis conditions, it is possible to acquire the atomic concentration distribution of chlorine atoms and nitrogen atoms at each position in the HAADF image. Specifically, the acquisition of the concentration distribution is performed by the following procedure.

The value of the carbon layer portion for FIB processing on the side of the magnetic layer 43 in the measurement region R is set as the background of the number of counts of each element, such as the number of Cl counts and the number of N counts, and the portion exceeding the counts is set as the surface of the magnetic layer 43. Line analysis is performed from the magnetic surface side in the measurement region R in 2.1 nm increments with the moving average of 6.3 nm horizontal to the magnetic surface, the number of Cl counts and the number of N counts are obtained for each line and quantitated using ZAF correction, and the atomic concentration [atm %] of each of Cl and N when the total number of atoms of the detected elements is 100% is obtained. The maximum atomic concentrations A1 and A2 [atm %] of Cl and N are obtained within the range of 150 nm from the magnetic surface, and the atomic concentration ratio (A2/A1) thereof is calculated. FIG. 11 shows an example of the measurement results of atomic concentration distributions of Cl and N. In FIG. 11, a peak P_A indicates the peak of the Cl concentration distribution and a peak P_B indicates the peak of the N concentration distribution. The maximum atomic concentrations A1 and A2

respectively mean the maximum values of atomic concentrations (acquired data) of the portions of the peak P_A and the peak P_B .

The ZAF correction is correction taking into account the influence of the following three effects, i.e., the atomic number effect, the absorption effect, and the fluorescence excitation effect, on the characteristic X-ray intensity.

Atomic number effect: the penetration depth of the electron beam changes depending on the constituent element of the sample or the concentration.

Absorption effect: the ease with which the generated characteristic X-rays escape from the sample changes depending on the constituent element or the concentration.

Fluorescence excitation effect: some characteristic X-rays are generated by characteristic X-rays emitted from other elements and continuous X-rays, and the amount of characteristic X-rays changes depending on the sample.

(Maximum Chlorine Atomic Concentration A1 of Magnetic Layer)

The maximum chlorine atomic concentration A1 obtained by measuring the chlorine atomic concentration of the magnetic layer 43 in the thickness direction of the magnetic layer 43 is favorably 4.5 atm % or more, 6.5 atm %. When the maximum chlorine atomic concentration A1 is 4.5 atm % or more, it is possible to prevent the content of the vinyl chloride resin in the magnetic layer 43 from becoming too small. This makes it possible to prevent the surface of the magnetic layer 43 from becoming too soft and prevent the protrusion 430 on the magnetic surface by travelling of the magnetic tape MT from being plastically deformed. Meanwhile, in the case where the maximum chlorine atomic concentration A1 is 6.5 atm % or less, it is possible to prevent the content of the vinyl chloride resin in the magnetic layer 43 from becoming too high. This makes it possible to prevent the surface of the magnetic layer 43 from becoming too hard and prevent the surface of the magnetic layer 43 from being subjected to brittle fracture when the head unit 56 slides over the magnetic tape MT.

The maximum chlorine atomic concentration A1 is obtained in the same procedure as that for the maximum chlorine atomic concentration A1 in the above method of measuring the atomic concentration ratio (A2/A1). (Average Value of Protruding Peak Height Rpk)

The average value of protruding peak heights Rpk on the surface of the magnetic layer 43 is 2.10 nm or less, favorably 1.84 nm or less. When the average value of the protruding peak heights Rpk exceeds 2.10 nm, there is a possibility that the surface region of the magnetic layer 43 that comes into contact with the head unit 56 and is subjected to initial wear becomes larger. Further, when the average value of the protruding peak heights Rpk exceeds 2.10 nm, there is a possibility that the height of the protrusion 430 on the surface of the magnetic layer 43 is too high, the spacing with the head unit 56 increases, and the electromagnetic conversion characteristics deteriorate.

The average value of the protruding peak heights Rpk is obtained as follows. First, the magnetic tape MT housed in the cartridge 10 is unwound, and the magnetic tape MT is cut into a length of 100 mm at a position 20 m in the longitudinal direction from the connection part 21 between the magnetic tape MT and the leader tape LT to obtain a sample. Next, the surface shape of the sample on the side of the magnetic layer 43 is measured at five positions in the longitudinal direction of the sample, the protruding peak heights Rpk obtained at the respective positions are simply averaged (arithmetically averaged) to obtain an average value, and the obtained

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average values is used as the average value of the protruding peak heights R_{pk} on the surface of the magnetic layer **43**.

The protruding peak height R_{pk} at each of the above positions is obtained as follows. First, the sample is placed on a slide glass such that the surface of the sample on the side of the magnetic layer **43** faces upward, and the end portion of the sample is fixed with mending tape. Next, the surface shape of the magnetic layer **43** is measured to obtain a corrected profile.

The measuring apparatus of the surface shape and the measurement conditions are as follows.

Apparatus: non-contact profilometer using optical interference

(non-contact surface/layer cross-sectional shape measurement system VertScan R5500GL-M100-AC manufactured by Ryoka Systems Inc.)

Objective lens: 50×

CCD: $\frac{1}{3}$ lens

Measurement region: 640×480 pixels (field of view: approximately 95 μm ×71 μm)

Measurement mode: phase

Wavelength filter: 520 nm

Noise removal filter: smoothing 3×3

Surface correction: correction using a quadratic polynomial approximation surface

Measurement software: VS-Measure Version 5.5.2

Analysis software: VS-viewer Version 5.5.5

As described above, the bearing of the analysis software is designated for the corrected profile to obtain the value of R_{pk} . For parameter settings, Min and Max are selected for both the vertical axis and the horizontal axis and the number of divisions is 1000.

(Standard Deviation σPES)

In a full volume test up to 40 times in an environment of 26° C. and 80% RH, a standard deviation σPES of the PES values obtained from the servo signal corresponding to the servo pattern is favorably less than 50 nm, more favorably 40 nm or less, still more favorably 30 nm or less, and particularly favorably 25 nm or less. The full volume test is a test in which data is recorded over the entire length and entire surface of the magnetic tape MT and reproduced. The 40 full volume tests are tests for evaluating whether it is capable withstanding continuous travelling for approximately one month.

The PES (Position Error Signal) indicates the deviation (error) of the read position of a servo pattern in the width direction of the magnetic tape MT when the servo pattern is reproduced (read) by the recording/reproduction apparatus. The standard deviation σPES is related to the friction of the magnetic surface, and the standard deviation σPES tends to increase as the friction of the magnetic surface increases. Therefore, by measuring the standard deviation σPES , it is possible to determine an increase in the friction of the magnetic surface.

Further, in order to adjust the tension in the longitudinal direction of the magnetic tape MT with high accuracy, it is favorable that the linearity of the servo band when the servo pattern is read by the recording/reproduction apparatus is as high as possible, i.e., the standard deviation σPES of the PES values indicating the deviation (error) of the read position is as low as possible. When the standard deviation σPES of the PES values of the magnetic tape MT is a low value as described above, the linearity of the servo band SB is high and it is possible to adjust the tension of the magnetic tape MT with high accuracy.

Part A of FIG. **12** is a diagram showing a first example of the change in standard deviation σPES over time of the PES

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values associated with travelling of the magnetic tape MT. As shown in Part A of FIG. **12**, when the standard deviation σPES is kept less than 50 nm in the full volume test up to 40 times, the increase in the friction of the magnetic surface associated with traveling of the magnetic tape MT is suppressed and track misalignment is suppressed. Part B of FIG. **12** is a diagram showing a second example of the change in standard deviation σPES over time of the PES values associated with travelling of the magnetic tape MT. As shown in Part B of FIG. **12**, when the standard deviation σPES is 50 nm or more in the full volume test up to 40 times, the friction of the magnetic surface increases as the magnetic tape MT travels, track misalignment frequently occurs, and thus, there is a possibility that the travelling of the magnetic tape MT stops.

A head unit **300** for PES measurement, which is to be used for measuring the PES value, will be described with reference to FIG. **13**. As the head unit **300**, an LTO2 head (head conforming to the LTO2 standard) manufactured by HPE (Hewlett Packard Enterprise) is used. The head unit **300** includes two head portions **300A** and **300B** arranged side by side along the longitudinal direction of the magnetic tape MT. Each head portion includes a plurality of recording heads **340** for recording a data signal on the magnetic tape MT, a plurality of reproduction heads **350** for reproducing the data signal recorded on the magnetic tape MT, and a plurality of servo heads **320** for reproducing the servo signal recorded on the magnetic tape MT. Note that in the case where the head unit **300** is used to only measure the PES value, the head unit **300** does not necessarily need to include the recording heads **340** and the reproduction heads **350**.

A measurement method of the standard deviation σPES will be described with reference to FIG. **6** and FIG. **13**. The PES value is measured to obtain the standard deviation σPES . The above head unit **300** for PES measurement is used to measure the PES value. The standard deviation σPES is measured for each of the 1st to 40th full volume tests. The standard deviation σPES may be measured while the head unit **300** is maintained parallel to the axis Δx parallel to the width direction of the magnetic tape MT or while the head unit **300** is kept inclined with respect to the axis Ax parallel to the width direction of the magnetic tape MT.

The standard deviation σPES of the PES values is calculated using the servo signal that has been corrected for the movement of the magnetic tape MT in the lateral direction. Further, the servo signal is subjected to high pass filter processing in order to reflect the followability of the head unit **300**. In the present disclosure, the standard deviation σPES is obtained using the signal obtained by performing the above correction and the above high pass filter processing on the servo signal, and is so-called written in $\text{PES}\sigma$.

First, while performing a full volume test, the servo signal of the magnetic tape MT is read by the head unit **300**. More specifically, the magnetic tape MT is caused to travel at the speed of 2 m/s such that the servo head **320** of the head portion **300A** and the servo head **320** of the head portion **300B** sequentially face each servo pattern of a predetermined servo band. Then, the above two servo heads **320** read the servo pattern and output the read servo pattern as a servo signal. The reading of the servo signal is performed in an environment of 26° C. and 80% RH.

Next, a subtraction is performed in the signals acquired by the head portions **300A** and **300B** as shown in FIG. **14** to obtain servo signal that has been corrected for the movement of the magnetic tape MT in the lateral direction. Then, the corrected servo signal is subjected to high pass filter processing.

cessing. When the magnetic tape MT is actually caused to travel by a drive, the recording/reproducing head mounted on the drive is caused to move by an actuator so as to follow the servo signal in the width direction of the magnetic tape MT. Since the Written in PES σ is a noise value after taking into account the followability of this head in the width direction, the above high pass filter processing is necessary. Therefore, although the high pass filter is not particularly limited, it is necessary to use a function that is capable of reproducing the followability of the above drive head in the width direction.

Next, the signal obtained by the above high pass filter processing is used to calculate the PES value for each servo frame **110** shown in FIG. 6. The standard deviation (Written in PES σ) of PES values calculated over 1 m from the position 20 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT is the standard deviation σ PES of the PES values in the present disclosure. The standard deviation σ PES is measured each of the full volume tests as described above.

The value of the PES value for each servo frame **110** is calculated by the following formula.

$$PES[\mu m] = \frac{X[\mu m] - \frac{(B_{a1} - A_{a1}) + (B_{a2} - A_{a2}) + (B_{a3} - A_{a3}) + (B_{a4} - A_{a4}) + (D_{a1} - C_{a1}) + (D_{a2} - C_{a2}) + (D_{a3} - C_{a3}) + (D_{a4} - C_{a4})}{(C_{a1} - A_{a1}) + (C_{a2} - A_{a2}) + (C_{a3} - A_{a3}) + (C_{a4} - A_{a4}) + (A'_{a1} - C_{a1}) + (A'_{a2} - C_{a2}) + (A'_{a3} - C_{a3}) + (A'_{a4} - C_{a4})} \times Y[\mu m]}{2 \times \tan \varphi} \quad (\text{Math. 3})$$

Here, the center line shown in FIG. 6 is the center line of the servo band.

X [μ m] represents the distance between a servo pattern A1 and a servo pattern B1 on the center line shown in FIG. 6, and Y [μ m] represents the distance between the servo pattern A1 and a servo pattern C1 on the center line shown in FIG. 6. X and Y are obtained by developing the magnetic tape MT with a ferricolloid developer after 40 full volume tests and using a universal tool microscope (TOPCON TUM-220ES) and a data processing apparatus (TOPCON CA-1B). Fifty servo frames are selected at a position 20 m in the longitudinal direction from the connection part **21** between the magnetic tape MT and the leader tape LT, X and Y are obtained in each servo frame, and the 50 pieces of data are simply averaged to be used as X and Y, which are to be substituted into the above formula.

The above difference (Ba1-Aa1) represents the time [sec] on the actual path between the two corresponding servo pattern B1 and servo pattern A1. Similarly, each of the other difference terms represents the time [sec] on the actual path between the two corresponding servo patterns. These times are obtained from the time between timing signals obtained from the waveform of the servo signal and the tape travelling speed. In the present specification, the actual path means the position where the servo head **320** that reads a servo signal actually travels on the servo signal. φ represents the average value of azimuth angles θ_1 and θ_2 (see FIG. 13), and is specifically obtained from $(\theta_1 + \theta_2)/2$. θ_1 and θ_2 are obtained by developing the magnetic tape MT with a ferricolloid developer after 40 full volume tests and using a universal tool microscope (TOPCON TUM-220ES) and a data processing apparatus (TOPCON CA-1B), similarly to the above measurement of the distances X and Y. Note that although the values of the difference (Ba1-Aa1) and the other differences are values obtained in each of the 40 full volume

tests, the distances X and Y and the azimuth angles θ_1 and θ_2 are values only once after the 40 full volume tests.

4 Method of Producing Magnetic Tape

Next, an example of the method of producing the magnetic tape MT having the above configuration will be described.

(Process of Preparing Paint)

First, non-magnetic particles, a binder, and the like are kneaded and dispersed in a solvent to prepare a coating material for forming an underlayer. Next, magnetic particles, the first particle **431**, the second particle **432**, a binder, and the like are kneaded and dispersed in a solvent to prepare a coating material for forming a magnetic layer. For preparing the coating material for forming a magnetic layer and coating material for forming an underlayer, for example, the following solvent, dispersing apparatus, and kneading apparatus can be used.

Examples of the solvent to be used for preparing the above paint include a ketone solvent such as acetone, methyl ethyl ketone, methyl isobutyl ketone, and cyclohexanone, an

alcohol solvent such as methanol, ethanol, and propanol, an ester solvent such as methyl acetate, ethyl acetate, butyl acetate, propyl acetate, ethyl lactate, and ethylene glycol acetate, an ether solvent such as diethylene glycol dimethyl ether, 2-ethoxyethanol, tetrahydrofuran, and dioxane, an aromatic hydrocarbon solvent such as benzene, toluene, and xylene, and a halogenated hydrocarbon solvent such as methylene chloride, ethylene chloride, carbon tetrachloride, chloroform, and chlorobenzene. These may be used alone or may be appropriately mixed and used.

Examples of the kneading apparatus to be used for preparing the above paint include, but not particularly limited to, a kneading apparatus such as a continuous twin-screw kneader, a continuous twin-screw kneader capable of performing dilution in multiple stages, a kneader, a pressure kneader, and a roll kneader. Further, examples of the dispersing apparatus to be used for preparing the above paint include, but particularly not limited to, a dispersing apparatus such as a roll mill, a ball mill, a horizontal sand mill, a perpendicular sand mill, a spike mill, a pin mill, a tower mill, a pearl mill (e.g., "DCP mill" manufactured by Eirich Co., Ltd.), a homogenizer, and an ultrasonic disperser.

(Application Process)

Next, the coating material for forming an underlayer is applied to one main surface of the base **41** and dried to form the underlayer **42**. Subsequently, the coating material for forming a magnetic layer is applied to this underlayer **42** and dried to form the magnetic layer **43** on the underlayer **42**. Note that during drying, the magnetic field of the magnetic particles is oriented in the thickness direction of the base **41** by, for example, a solenoid coil. Further, during drying, the magnetic field of the magnetic particles may be directed in the travelling direction (longitudinal direction) of the base **41** and then directed in the thickness direction of the base **41** by, for example, a solenoid coil. By performing processing

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of orienting the magnetic particles in the longitudinal direction in this way, it is possible to further improve the degree of perpendicular orientation (i.e., the squareness ratio S1) of the magnetic particles. After forming the magnetic layer 43, the back layer 44 is formed on the other main surface of the base 41. In this way, the magnetic tape MT is obtained.

The squareness ratios S1 and S2 are set to desired values by adjusting, for example, the strength of the magnetic field to be applied to the coating film of the coating material for forming a magnetic layer, the concentration of the solid content in the coating material for forming a magnetic layer, and the drying conditions (drying temperature and drying time) of the coating film of the coating material for forming a magnetic layer. The strength of the magnetic field to be applied to the coating film is favorably two or more times and three or less times the coercive force of the magnetic particles. In order to further increase the squareness ratio S1 (i.e., in order to further decrease the squareness ratio S2), it is favorable to improve the dispersion state of the magnetic particles in the coating material for forming a magnetic layer. Further, in order to further increase the squareness ratio S1, it is also effective to magnetize the magnetic particles before the coating material for forming a magnetic layer enters an orientation device for orienting the magnetic field of the magnetic particles. Note that the above methods of adjusting the squareness ratios S1 and S2 may be used alone, or two or more of them may be used in combination. (Curing Process)

Next, the magnetic tape MT is wound into a roll, and then, the magnetic tape MT in this state is subjected to heat treatment to cure the underlayer 42 and the magnetic layer 43.

(Calendering Process)

Next, the obtained magnetic tape MT is calendered to smooth the magnetic surface.

(Demagnetizing Process and Process of Writing Servo Pattern)

Next, as necessary, the magnetic tape MT may be demagnetized, and then, a servo pattern may be written to the magnetic tape MT.

(Cutting Process)

Next, the magnetic tape MT is cut into a predetermined width (e.g., 1/2 inch width). In this way, the magnetic tape MT is obtained.

(Method of Adjusting Average Value of Protruding Peak Heights Rpk)

The average value of the protruding peak heights Rpk on the surface of the magnetic layer 43 can be adjusted to a desired value by adjusting, for example, the thickness of the magnetic layer 43, the surface properties of the base 41, the surface properties of the underlayer 42, the particle size of the inorganic additive to be blended in the coating material for forming a magnetic layer, or the dispersion time of the coating material for forming a magnetic layer. The average value of the protruding peak heights Rpk may be adjusted by combining two or more of the above conditions. Specifically, for example, by reducing the thickness of the magnetic layer 43, the average value of the protruding peak heights Rpk can be increased. By roughening the surface properties of the base 41 and/or the underlayer 42, the average value of the protruding peak heights Rpk can be increased. The surface properties of the underlayer 42 can be adjusted by, for example, adjusting the amount of carbon added and/or the dispersion time. In the present specification, "and/or" means at least one. For example, in the case of "X and/or Y", there are three meanings: only X, only Y, and X and Y.

(Method of Adjusting Atomic Concentration Ratio (A2/A1))

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The atomic concentration ratio (A2/A1) can be adjusted to a desired value by, for example, adjusting the amounts of the urethane resin, the vinyl chloride resin, and the curing agent to be blended in the coating material for forming a magnetic layer. The atomic concentration ratio (A2/A1) can also be adjusted to a desired value by, for example, adjusting the solid content of the coating material for forming a magnetic layer and/or the blending ratio of the non-magnetic particles in the coating material for forming an underlayer and the binder. Specifically, for example, when the solid content of the coating material for forming a magnetic layer is high, the amount of solvent that penetrates into the underlayer 42 is reduced, the amount of binder that moves to the underlayer 42 is reduced, and thus, the atomic concentration ratio (A2/A1) changes. By adjusting the blending ratio of the non-magnetic particles in the coating material for forming an underlayer and the binder to increase the amount of binder that is not adsorbed on the non-magnetic particles, the amount of binder that moves to the magnetic layer 43 increases and the atomic concentration ratio (A2/A1) changes.

5 Operation and Effect

As described above, in the magnetic tape MT according to an embodiment, the atomic concentration ratio (A2/A1) of the maximum chlorine atomic concentration A1 obtained by measuring the chlorine atomic concentration of the magnetic layer 43 in the thickness direction of the magnetic layer 43 and the maximum nitrogen atomic concentration A2 obtained by measuring the nitrogen atomic concentration of the magnetic layer 43 in the thickness direction of the magnetic layer 43 is 0.50 or more and 0.75 or less. This allows the hardness of the surface of the magnetic layer 43 to be adjusted within a predetermined range. Therefore, it is possible to suppress brittle fracture of the protrusion 430 on the surface of the magnetic layer 43 and prevent the protrusion 430 on the surface of the magnetic layer 43 from being plastically deformed. Therefore, it is possible to suppress an increase in friction between the magnetic tape MT and the head unit 56 during recording or reproduction.

Further, in the magnetic tape MT according to an embodiment, the average value of the protruding peak heights Rpk on the surface of the magnetic layer 43 is 2.10 nm or less. This reduces the spacing with the head unit 56 and provides favorable electromagnetic conversion characteristics. Further, it is possible to reduce the surface region of the magnetic layer 43 that comes into contact with the head unit 56 and is subjected to initial wear.

In the case where the servo stripes 113 of the A burst 111A and the servo stripes 113 of the B burst 111B are asymmetric and the servo stripes 113 of the C burst 112C and the servo stripes 113 of the D burst 112D are asymmetric as shown in FIG. 6, assumption is made that the head unit 56 is kept inclined with respect to the axis Ax parallel to the width direction of the magnetic tape MT during recording and reproduction of the magnetic tape MT.

As shown in FIG. 15, when recording and reproduction of an existing magnetic tape MT1 are performed while the head unit 56 is kept inclined as described above, the surface protrusion is scraped off at the end portion on the magnetic surface in the width direction of the magnetic tape MT1, a scraped powder 57 falls off, and the scraped powder 57 spreads over the entire surface of the head unit 56 due to travelling of the magnetic tape MT1. For this reason, there is a possibility that the friction between the magnetic tape

MT1 and the head unit 56 increases during recording or reproduction and travelling stability is reduced.

On the other hand, in the magnetic tape MT according to an embodiment, since brittle fracture of the magnetic layer 43 can be suppressed as described above, it is possible to suppress the increase in friction between the magnetic tape MT and the head unit 56 even in the case where the head unit 56 is kept inclined during recording or reproduction and suppress a decrease in travelling stability.

6 Modified Example

Although the case where the magnetic tape cartridge is the one-reel type cartridge 10 has been described in the above one embodiment, the magnetic tape cartridge may be a two-reel type cartridge.

FIG. 16 is an exploded perspective view showing an example of a configuration of a cartridge 121 of the two-reel type. The cartridge 121 includes an upper half 102 formed of a synthetic resin, a transparent window member 123 that is fitted into and fixed to a window portion 102a that is opened on the upper surface of the upper half 102, a reel holder 122 that is fixed to the inside of the upper half 102 and prevents reels 106 and 107 from floating, a lower half 105 corresponding to the upper half 102, the reels 106 and 107 housed in the space formed by combining the upper half 102 and the lower half 105, the magnetic tape MT wound on the reels 106 and 107, a front lid 109 that closes a front-side opening formed by combining the upper half 102 and the lower half 105, and a back lid 109A that protects the magnetic tape MT exposed in the front-side opening.

The reels 106 and 107 are for winding the magnetic tape MT. The reel 106 includes a lower flange 106b that includes a cylindrical hub portion 106a in the center around which the magnetic tape MT is wound, an upper flange 106c having substantially the same size as the lower flange 106b, and a reel plate 108 sandwiched between the hub portion 106a and the upper flange 106c. The reel 107 has a configuration similar to that of the reel 106.

The window member 123 is provided with mounting holes 123a for mounting the reel holders 122 at positions corresponding to the reels 106 and 107, the reel holders 122 being reel holding means that prevent these reels from floating. The magnetic tape MT is similar to the magnetic tape MT according to the first embodiment.

EXAMPLES

Although the present disclosure will be specifically described with reference to examples, the present disclosure is not limited to these Examples.

In the following Examples and Comparative Examples, the average aspect ratio of the magnetic particles, the average particle volume of the magnetic particles, the average thickness of the magnetic tape, the average thickness of the magnetic layer, the average thickness of the underlayer, the average thickness of the back layer, the maximum chlorine atomic concentration A1 of the magnetic layer, the maximum nitrogen atomic concentration A2 of the magnetic layer, the atomic concentration ratio (A2/A1) of the magnetic layer, the average value of the protruding peak heights Rpk of the surface of the magnetic layer, the squareness ratio S1 of the magnetic layer in the perpendicular direction of the magnetic tape, and the squareness ratio S2 of the magnetic layer in the longitudinal direction of the magnetic tape are values obtained by the measurement methods described in the above one embodiment.

Example 1

(Process of Preparing Coating Material for Forming Magnetic Layer)

A coating material for forming a magnetic layer was prepared as follows. First, a first composition of the following formulation was kneaded with an extruder. Next, the kneaded first composition and a second composition of the following formulation were added to a stirring tank including a dispersion device to perform preliminary mixing. Subsequently, dyno mill mixing was further performed and filter treatment was performed to prepare a coating material for forming a magnetic layer.

(First Composition)

Barium ferrite ($\text{BaFe}_{12}\text{O}_{19}$) magnetic powder (hexagonal plate shape, average aspect ratio of 3.0, average particle volume of 1200 nm^3): 100 parts by mass

Vinyl chloride resin solution (formulation of the resin solution: 30 mass % of a vinyl chloride resin, 70 mass % of a cyclohexanone solution): 35 parts by mass

(the degree of polymerization of 300, $\text{Mn}=10000$, including $\text{OSO}_3\text{K}=0.07 \text{ mmol/g}$, secondary $\text{OH}=0.3 \text{ mmol/g}$ as polar groups.)

Polyurethane resin solution (amount of the blended resin solution: the amount of blended polyurethane resin 30 mass %, the amount of blended cyclohexanone 70 mass %): 15 parts by mass

(polyurethane resin: number average molecular weight $\text{Mn}=25000$, $\text{Tg}=110^\circ \text{ C.}$)

Aluminum oxide powder: 4 parts by mass ($\alpha\text{-Al}_2\text{O}_3$, average particle size of $0.1 \mu\text{m}$)

(Second Composition)

Carbon black: 2 parts by mass (manufactured by Tokai Carbon Co., Ltd., product name: SEAST TA)

Polyurethane resin solution (formulation of the resin solution: the amount of blended polyurethane resin 30 mass %, cyclohexanone 70 mass %): 5.56 parts by mass

(polyurethane resin: number average molecular weight $\text{Mn}=25000$, $\text{Tg}=110^\circ \text{ C.}$)

n-butyl stearate: 2 parts by mass

Methyl ethyl ketone: 121.0 parts by mass

Toluene: 121.0 parts by mass

Cyclohexanone: 116.0 parts by mass

Finally, a polyisocyanate (product name: Coronate L, manufactured by TOSOH CORPORATION): 3.3 parts by mass and stearic acid: 2 parts by mass were added as curing agents to the coating material for forming a magnetic layer prepared as described above.

(Process of Preparing Coating Material for Forming Underlayer)

A coating material for forming an underlayer was prepared as follows. First, a third composition of the following formulation was kneaded with an extruder. Next, the kneaded third composition and a fourth composition of the following formulation were added to a stirring tank including a dispersion device to perform preliminary mixing. Subsequently, dyno mill mixing was further performed and filter treatment was performed to prepare a coating material for forming an underlayer.

(Third Composition)

Acicular iron oxide powder: 100 parts by mass ($\alpha\text{-Fe}_2\text{O}_3$, average major axis length of $0.11 \mu\text{m}$)

Vinyl chloride resin solution (formulation of the resin solution: 30 mass % of a vinyl chloride resin, 70 mass % of cyclohexanone): 46 parts by mass

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(the degree of polymerization of 300, Mn=10000, including OSO_3K =0.07 mmol/g, secondary OH=0.3 mmol/g as polar groups.)

(average particle size of 20 nm)

(Fourth Composition)

Carbon black: 35 parts by mass

Polyurethane resin solution (formulation of the resin solution: 30 mass % of polyurethane resin, 70 mass % of cyclohexanone): 50 parts by mass

(polyurethane resin: number average molecular weight Mn=25000, glass transition temperature T_g =70° C.)

n-butyl stearate: 2 parts by mass

Methyl ethyl ketone: 108.2 parts by mass

Toluene: 108.2 parts by mass

Cyclohexanone: 100.0 parts by mass

Finally, a polyisocyanate (product name: Coronate L, manufactured by TOSOH CORPORATION): 2.49 parts by mass and stearic acid: 2 parts by mass were added as curing agents to the coating material for forming an underlayer prepared as described above.

(Process of Preparing Coating Material for Forming Back Layer)

A coating material for forming a back layer was prepared as follows. A coating material for forming a back layer was prepared by mixing the following raw material in a stirring tank including a dispersion device and performing filter treatment thereon.

Carbon black (manufactured by ASAHI CARBON CO., LTD., product name: #80): 100 parts by mass

Polyester polyurethane: 100 parts by mass

(manufactured by Nippon Polyurethane Co., Ltd., product name: N-2304)

Methyl ethyl ketone: 500 parts by mass

Toluene: 400 parts by mass

Cyclohexanone: 100 parts by mass

Polyisocyanate (Product name: Coronate L, manufactured by TOSOH CORPORATION): 10 parts by mass

(Application Process)

The coating material for forming a magnetic layer and coating material for forming an underlayer prepared as described above were used to form an underlayer and a magnetic layer on one main surface of a long polyethylene naphthalate film (hereinafter, referred to as a "PEN film") having an average thickness of 3.60 μm , which is a non-magnetic support, as follows. First, the coating material for forming an underlayer was applied to the one main surface of the PEN film and dried to form an under layer such that the average thickness is 1070 nm after calendaring. Next, the coating material for forming a magnetic layer was applied to the underlayer and dried to form a magnetic layer such that the average thickness is 80 nm after calendaring. Note that during drying of the coating material for forming a magnetic layer, the magnetic field of the magnetic particles was oriented in the thickness direction of the film by a solenoid coil. Further, the squareness ratio S1 in the perpendicular direction (thickness direction) of the magnetic tape was set to 65% and the squareness ratio S2 in the longitudinal direction of the magnetic tape was set to 38%. Subsequently, the coating material for forming a back layer was applied to the other main surface of the PEN film and dried to form a back layer such that the average thickness is 0.45 μm after calendaring. In this way, a magnetic tape was obtained.

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(Curing Process)

The magnetic tape was wound into a roll, and then, the magnetic tape in this state was subjected to heat treatment at 70° C. for 48 hours to cure the underlayer and the magnetic layer.

(Calendaring Process)

Calendaring was performed to smooth the surface of the magnetic layer. At this time, the calendaring temperature was set to 100° C. and the calendaring pressure was set to 200 kg/cm.

(Cutting Process)

The magnetic tape obtained as described above was cut into a 1/2 inch (12.65 mm) width. In this way, a magnetic tape in which the atomic concentration ratio (A_2/A_1) was 0.71, the average value of the protruding peak heights Rpk of the surface of the magnetic layer was 1.64, and the average thickness was 5.20 μm was obtained.

Example 2

The amount of carbon black blended in the second composition was changed to 2.5 parts by mass (manufactured by Tokai Carbon Co., Ltd.: SEAST TA) and the average thickness of the underlayer after calendaring was changed to 670 nm. The average thickness of the PEN film as a base was changed to 4.20 μm and the average thickness of the back layer after calendaring was changed to 0.30 μm . A magnetic tape in which the atomic concentration ratio (A_2/A_1) was 0.71, the average value of the protruding peak heights Rpk was 1.80, and the average thickness was 5.25 μm was obtained in the same manner as in Example 1 except for the above.

Example 3

The amount of the vinyl chloride resin solution blended in the first composition was changed to 40 parts by mass, and the amount of the polyurethane resin solution blended in the first composition was changed to 10 parts by mass. Further, the average thickness of the PEN film as a base was changed to 4.00 μm , the average thickness of the magnetic layer after calendaring was changed to 70 nm, the average thickness of the underlayer after calendaring was changed to 880 nm, and the average thickness of the back layer after calendaring was changed to 0.30 μm . A magnetic tape in which the atomic concentration ratio (A_2/A_1) was 0.55, the average value of the protruding peak heights Rpk was 1.68, and the average thickness was 5.25 μm was obtained in the same manner as in Example 1 except for the above.

Example 4

The amount of the vinyl chloride resin solution blended in the first composition was changed to 40 parts by mass and the amount of the polyurethane resin solution blended in the first composition was changed to 10 parts by mass. Further, the average thickness of the PEN film as a base was changed to 4.10 μm , the average thickness of the magnetic layer after calendaring was changed to 60 nm, the average thickness of the underlayer after calendaring was changed to 600 nm, and the average thickness of the back layer after calendaring was changed to 0.25 μm . A magnetic tape in which the atomic concentration ratio (A_2/A_1) was 0.55, the average value of the protruding peak heights Rpk was 1.84, and the average thickness was 5.01 μm was obtained in the same manner as in Example 1 except for the above.

Comparative Example 1

The amount of the vinyl chloride resin solution blended in the first composition was changed to 43 parts by mass and

the amount of the polyurethane resin solution blended in the first composition was changed to 7 parts by mass. Further, the average thickness of the PEN film as a base was changed to 4.00 μm , the average thickness of the magnetic layer after calendaring was changed to 70 nm, the average thickness of the underlayer after calendaring was changed to 880 nm, and the average thickness of the back layer after calendaring was changed to 0.30 μm . A magnetic tape in which the atomic concentration ratio (A2/A1) was 0.47, the average value of the protruding peak heights Rpk was 1.71, and the average thickness was 5.25 μm was obtained in the same manner as in Example 1 except for the above.

Comparative Example 2

The amount of the vinyl chloride resin solution blended in the first composition was changed to 30 parts by mass, the amount of the polyurethane resin solution blended in the first composition was changed to 20 parts by mass, and the amount of carbon black blended in the second composition was changed to 3.0 parts by mass (manufactured by Tokai Carbon Co., Ltd.: SEAST TA). Further, the average thickness of the PEN film as a base was changed to 4.00 μm , the average thickness of the underlayer after calendaring was changed to 1120 nm, and the average thickness of the back layer after calendaring was changed to 0.30 μm . A magnetic tape in which the atomic concentration ratio (A2/A1) was 0.76, the average value of the protruding peak heights Rpk was 2.14, and the average thickness was 5.50 μm was obtained in the same manner as in Example 1 except for the above.

Comparative Example 3

The amount of the vinyl chloride resin solution blended in the first composition was changed to 43 parts by mass, the amount of the polyurethane resin solution blended in the second composition was changed to 7 parts by mass, and the amount of carbon black blended in the second composition was changed to 3.5 parts by mass (manufactured by Tokai Carbon Co., Ltd.: SEAST TA). Further, the average thickness of the underlayer after calendaring was changed to 1120 nm and the average thickness of the back layer after calendaring was changed to 0.40 μm . A magnetic tape in which the atomic concentration ratio (A2/A1) was 0.47, the average value of the protruding peak heights Rpk was 2.47, and the average thickness was 5.20 μm was obtained in the same manner as in Example 1 except for the above.

Comparative Example 4

The amount of the vinyl chloride resin solution blended in the first composition was changed to 37 parts by mass, the amount of the polyurethane resin solution blended in the second composition was changed to 13 parts by mass, and the amount of carbon black blended in the second composition was changed to 3.5 parts by mass (manufactured by Tokai Carbon Co., Ltd.: SEAST TA). Further, the average thickness of the PEN film as a base was changed to 4.00 μm , the average thickness of the magnetic layer after calendaring was changed to 70 nm, the average thickness of the underlayer after calendaring was changed to 880 nm, and the average thickness of the back layer after calendaring was changed to 0.30 μm . A magnetic tape in which the atomic

concentration ratio (A2/A1) was 0.62, the average value of the protruding peak heights Rpk was 2.50, and the average thickness was 5.25 μm was obtained in the same manner as in Example 1 except for the above.

(Evaluation of Standard Deviation σPES)

First, a full volume test (entire-length and entire-surface recording and reproduction test) was repeated up to 40 times, and the number of full volume tests at which the standard deviation σPES was 50 nm or more was measured. The standard deviation σPES was measured by the method of measuring the standard deviation σPES described in the above one embodiment. However, the measurement of the standard deviation σPES was performed while the head unit is kept parallel to the axis Ax parallel to the width direction of the magnetic tape. Further, in the case where the standard deviation σPES reached 50 nm or more within 40 full volume tests, the distances X and Y and the azimuth angles θ_1 and θ_2 to be used for calculating the standard deviation σPES were measured at the time point when the standard deviation σPES reached 50 nm or more. Next, the standard deviation σPES was evaluated on the basis of the following criteria. The evaluation results are shown in Table 1.

Pass: the standard deviation σPES is less than 50 nm up to 40 full volume tests.

Fail: the standard deviation σPES reaches 50 nm or more within 40 full volume tests.

Note that in the case where the evaluation result was fail, the number of full volume tests at which the standard deviation σPES reached 50 nm or more was described in Table 1.

As described above, when the standard deviation σPES reaches 50 nm or more, since the friction of the magnetic surface increases and track misalignment frequently occurs, there is a possibility that travelling of the magnetic tape stops.

Further, as described above, the 40 full volume tests are each a test for evaluating whether it is capable withstanding continuous travelling for approximately one month.

(Evaluation of Electromagnetic Conversion Characteristics)

First, a reproduction signal of the magnetic tape was obtained using a loop tester (manufactured by Microphysics). The conditions for acquiring the reproduction signal are shown below.

Head: GMR head

Speed: 1.85 m/s

Signal: single recording frequency 10 MHz (as a 2T half Nyquist frequency)

Recording current: optimal recording current

Next, the reproduction signal was captured by a spectrum analyzer (spectrum analyzer) with a span of 0 to 20 MHz (resolution band width=100 kHz, VBW=30 kHz). Next, the peak of the captured spectrum was taken as a signal amount S, the floor noise excluding the peak was integrated from 3 MHz to 20 MHz to obtain a noise amount N, and a ratio S/N of the signal amount S and the noise amount N was obtained as an SNR (Signal-to-Noise Ratio). Next, the obtained SNR was converted into a relative value (dB) with reference to the SNR in Comparative Example 2 as a reference medium. The results are shown in Table 1.

TABLE 1

	Average thickness of magnetic layer [nm]	Average thickness of underlayer [nm]	Sum of average thicknesses of magnetic layer and underlayer [nm]	Average thickness of base [μm]	Average thickness of back layer [μm]	Average thickness of magnetic tape [μm]	Maximum Cl atomic concentration in magnetic layer A1 [atm %]
Example 1	80	1070	1150	3.60	0.45	5.20	4.50
Example 2	80	670	750	4.20	0.30	5.25	4.50
Example 3	70	880	950	4.00	0.30	5.25	6.00
Example 4	60	600	660	4.10	0.25	5.01	6.00
Comparative Example 1	70	880	950	4.00	0.30	5.25	6.00
Comparative Example 2	80	1120	1200	4.00	0.30	5.50	4.20
Comparative Example 3	80	1120	1200	3.60	0.40	5.20	6.00
Comparative Example 4	70	880	950	4.00	0.30	5.25	4.50

	Maximum N atomic concentration in magnetic layer A2 [atm %]	Atomic concentration ratio (A2/A1)	Rpk average value [nm]	26° C., 80% Full volume test (Increase in σPES)	Electromagnetic conversion characteristics [dB]
Example 1	3.20	0.71	1.64	Pass (40FV)	1.2
Example 2	3.20	0.71	1.80	Pass (40FV)	0.7
Example 3	3.30	0.55	1.68	Pass (40FV)	1.0
Example 4	3.30	0.55	1.84	Pass (40FV)	0.9
Comparative Example 1	2.80	0.47	1.71	Fail (28FV)	0.8
Comparative Example 2	3.20	0.76	2.14	Fail (23FV)	0.0
Comparative Example 3	2.80	0.47	2.47	Fail (26FV)	-0.5
Comparative Example 4	2.80	0.62	2.50	Pass (40FV)	-0.2

In Table 1, Pass (40FV) represents that the standard deviation σPES was less than 50 nm up to 40 full volume tests.

Fail (23FV), Fail (26FV), and Fail (28FV) respectively represent that the standard deviation σPES reached 50 nm or more at the 23rd, 26th, and 28th full volume tests.

The following can be seen from Table 1.

When the atomic concentration ratio (A2/A1) of the magnetic layer is 0.50 or more and 0.75 or less and the average value of the protruding peak heights Rpk of the surface of the magnetic layer is 2.10 nm or less, favorable electromagnetic conversion characteristics can be achieved and it is possible to suppress the standard deviation σPES to less than 50 nm up to 40 full volume tests. That is, it is capable of achieving favorable electromagnetic conversion characteristics and suppressing the increase in friction.

In Comparative Example 4, the average value of the protruding peak heights Rpk exceeds 2.10 nm, but the increase in the standard deviation σPES (i.e., friction) is suppressed. This is presumably because in Comparative Example 4, the atomic concentration ratio (A2/A1) of the magnetic layer is within the range of 0.50 or more and 0.75 or less and thus the protrusions are less likely to be scraped off due to initial wear and the amount of powder fall is reduced even if the average value of the protruding peak heights Rpk exceeds 2.10 nm.

Although embodiments and modified examples of the present disclosure have been specifically described above, the present disclosure is not limited to the above embodiments and modified examples, and various modifications can be made on the basis of the technical idea of the present disclosure. For example, the configurations, methods, processes, shapes, materials, numerical values, and the like mentioned in the above embodiments and modified examples are merely examples, and configurations, methods, processes, shapes, materials, numerical values, and the like different from these may be used as necessary. The configurations, methods, processes, shapes, materials, numerical values, and the like of the above embodiments

and modified examples can be combined with each other without departing from the essence of the present disclosure.

The chemical formulae of compounds and the like exemplified in the above embodiments and modified examples are representative ones, and they are not limited to the stated valances and the like as long as they are general names of the same compounds. In the numerical ranges described in stages in the above embodiments and modified examples, the upper limit value or the lower limit value in the numerical range of one stage may be replaced with the upper limit value or the lower limit value in the numerical range of another stage. The materials exemplified in the above embodiments and modified examples can be used alone, or two or more of them can be used in combination, unless otherwise specified.

Further, the present disclosure may also take the following configurations.

- (1) A tape-shaped magnetic recording medium, including:
 - a base; and
 - a magnetic layer,
 - an average thickness of the magnetic recording medium being 5.30 μm or less,
 - the magnetic layer containing a chlorine atom and a nitrogen atom,
 - an atomic concentration ratio (A2/A1) of a maximum chlorine atomic concentration A1 obtained by measuring a chlorine atomic concentration of the magnetic layer in a thickness direction of the magnetic layer and a maximum nitrogen atomic concentration A2 obtained by measuring a nitrogen atomic concentration of the magnetic layer in the thickness direction of the magnetic layer being 0.50 or more and 0.75 or less,
 - an average value of a protruding peak height Rpk of a surface of the magnetic layer being 2.10 nm or less.
- (2) The magnetic recording medium according to (1), in which
 - the magnetic layer has a servo pattern,
 - the servo pattern has a plurality of first magnetized regions and a plurality of second magnetized regions, and

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- the plurality of first magnetized regions and the plurality of second magnetized regions are asymmetric with respect to an axis parallel to a width direction of the magnetic recording medium.
- (3) The magnetic recording medium according to (2), in which
an inclination angle of each of the first magnetized regions with respect to the axis and an inclination angle of each of the second magnetized regions with respect to the axis are different from each other.
- (4) The magnetic recording medium according to (3), in which
of the inclination angle of each of the first magnetized regions the inclination angle of each of the second magnetized region, the larger inclination angle is 180° or more and 28° or less.
- (5) The magnetic recording medium according to any one of (2) to (4), in which
a standard deviation σ_{PES} of PES (Position Error Signal) values obtained from a servo signal corresponding to the servo pattern is less than 50 nm in a full volume test up to 40 times in an environment of 26° C. and 80% RH.
- (6) The magnetic recording medium according to any one of (1) to (5), in which
the maximum chlorine atomic concentration A1 is 4.5 atm % or more, 6.5 atm %.
- (7) The magnetic recording medium according to any one of (1) to (6), in which
the atomic concentration ratio (A1/A2) is 0.55 or more and 0.71 or less.
- (8) The magnetic recording medium according to any one of (1) to (7), in which
the magnetic layer includes a first binder and a second binder,
the first binder contains the chlorine atom, and
the second binder contains the nitrogen atom.
- (9) The magnetic recording medium according to any one of (1) to (7), in which
the magnetic layer includes a vinyl chloride resin and a urethane resin.
- (10) The magnetic recording medium according to any one of (1) to (9), in which
the magnetic layer includes a large number of protrusions on a surface thereof.
- (11) The magnetic recording medium according to (10), in which
the large number of protrusions are formed by an abrasive and a carbon particle.
- (12) The magnetic recording medium according to any one of (1) to (11), further including
an underlayer,
an average thickness of the underlayer being 900 nm or less.
- (13) The magnetic recording medium according to any one of (1) to (12), in which
an average thickness of the magnetic layer is 80 nm or less.
- (14) The magnetic recording medium according to any one of (1) to (13), in which
an average thickness of the base is 4.40 μm or less.
- (15) The magnetic recording medium according to any one of (1) to (14), in which
the magnetic layer includes a magnetic particle, and
the magnetic particle contains hexagonal ferrite, ϵ -iron oxide, or Co-containing spinel ferrite.

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- (16) A cartridge, including:
the magnetic recording medium according to any one of (1) to (15).

REFERENCE SIGNS LIST

10 cartridge
11 cartridge memory
31 antenna coil
32 rectification/power-supply circuit
33 clock circuit
34 detection/modulation circuit
35 controller
36 memory
36A first storage region
36B second storage region
41 base
42 underlayer
43 magnetic layer
44 back layer
56, 300 head unit
56A, 56B servo read head
110 servo frame
111 servo subframe 1
111A A burst
111B B burst
112 servo subframe 2
112C C burst
112D D burst
113 servo stripe
430 protrusion
431 first particle
432 second particle
MT magnetic tape
SB servo band
DB data bind

The invention claimed is:

1. A magnetic recording medium, comprising:
a base; and
a magnetic layer,
an average thickness of the magnetic recording medium being 5.30 μm or less,
the magnetic layer including a chlorine atom and a nitrogen atom,
an atomic concentration ratio (A2/A1) of a maximum chlorine atomic concentration A1 obtained by measuring a chlorine atomic concentration of the magnetic layer in a thickness direction of the magnetic layer and a maximum nitrogen atomic concentration A2 obtained by measuring a nitrogen atomic concentration of the magnetic layer in the thickness direction of the magnetic layer being 0.50 or more and 0.75 or less,
an average value of a protruding peak height Rpk of a surface of the magnetic layer being 2.10 nm or less,
wherein the magnetic recording medium is tape-shaped.
2. The magnetic recording medium according to claim 1,
wherein
the magnetic layer has a servo pattern,
the servo pattern has a plurality of first magnetized regions and a plurality of second magnetized regions,
and
the plurality of first magnetized regions and the plurality of second magnetized regions are asymmetric with respect to an axis parallel to a width direction of the magnetic recording medium.

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3. The magnetic recording medium according to claim 2, wherein
 an inclination angle of each of the first magnetized regions with respect to the axis and an inclination angle of each of the second magnetized regions with respect to the axis are different from each other.
4. The magnetic recording medium according to claim 3, wherein
 of the inclination angle of each of the first magnetized regions the inclination angle of each of the second magnetized region, the larger inclination angle is 18° or more and 28° or less.
5. The magnetic recording medium according to claim 2, wherein
 a standard deviation σ PES of PES (Position Error Signal) values obtained from a servo signal corresponding to the servo pattern is less than 50 nm in a full volume test up to 40 times in an environment of 26° C. and 80% RH.
6. The magnetic recording medium according to claim 1, wherein
 the maximum chlorine atomic concentration A1 is 4.5 atm % or more, 6.5 atm %.
7. The magnetic recording medium according to claim 1, wherein
 the atomic concentration ratio (A1/A2) is 0.55 or more and 0.71 or less.
8. The magnetic recording medium according to claim 1, wherein
 the magnetic layer includes a first binder and a second binder,
 the first binder includes the chlorine atom, and
 the second binder includes the nitrogen atom.

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9. The magnetic recording medium according to claim 1, wherein
 the magnetic layer includes a vinyl chloride resin and a urethane resin.
10. The magnetic recording medium according to claim 1, wherein
 the magnetic layer includes protrusions on a surface thereof.
11. The magnetic recording medium according to claim 10, wherein
 the protrusions are formed by an abrasive and a carbon particle.
12. The magnetic recording medium according to claim 1, further comprising
 an underlayer,
 an average thickness of the underlayer being 900 nm or less.
13. The magnetic recording medium according to claim 1, wherein
 an average thickness of the magnetic layer is 80 nm or less.
14. The magnetic recording medium according to claim 1, wherein
 an average thickness of the base is 4.40 μ m or less.
15. The magnetic recording medium according to claim 1, wherein
 the magnetic layer includes a magnetic particle, and
 the magnetic particle includes hexagonal ferrite, ϵ -iron oxide, or Co-containing spinel ferrite.
16. A cartridge, comprising:
 the magnetic recording medium according to claim 1.

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